Agile Process Development and Devops Lab
Assessment -2

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Registration No – 23MIS0074

Slot – L3+L4

Cours Code – ISWE406P

1. **Learning Goal** - Understand what Jenkins is and run a job successfully.

   Stepwise Tasks

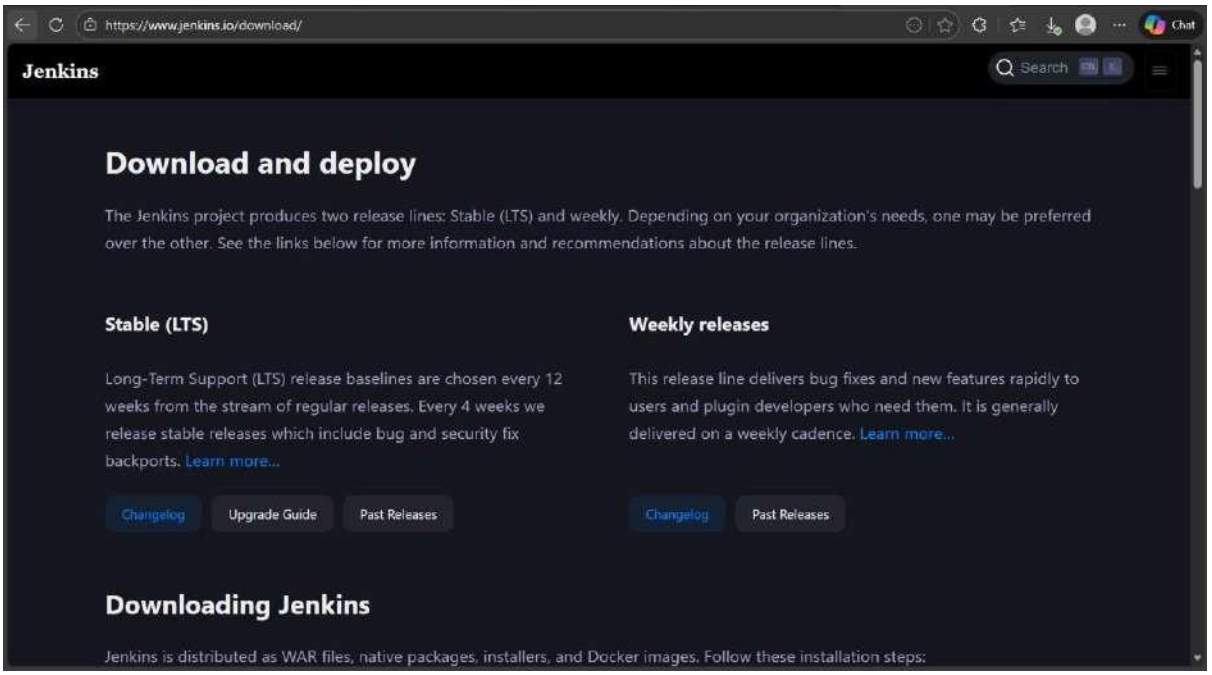
   1. **What is CI/CD? (Real-world example)**

      CI (Continuous Integration):
      Developers frequently merge code into a shared repository, and Jenkins automatically builds and tests it.
      CD (Continuous Delivery/Deployment):
      After a successful build, the application is automatically delivered or deployed.
      Real-world example:
      A team working on a shopping website pushes code to GitHub → Jenkins automatically builds the code → tests it → deploys the updated website without manual effort.

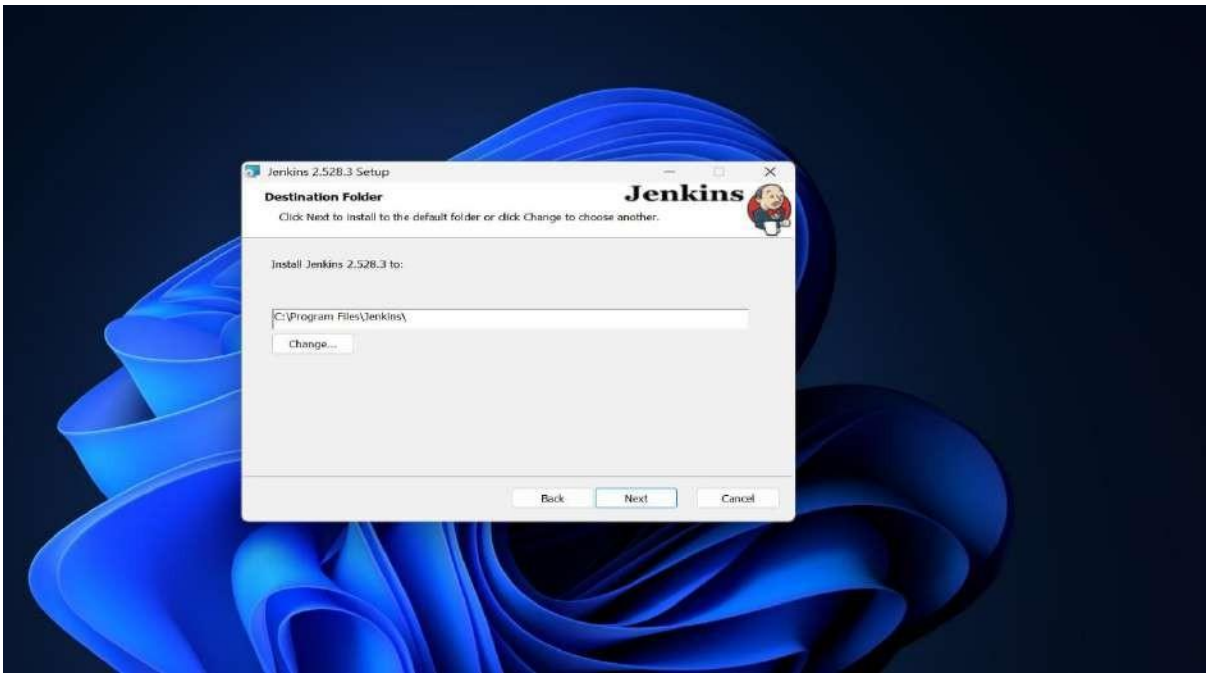
   2. **Install Jenkins (Windows / Linux)**

      Open jenkins.io

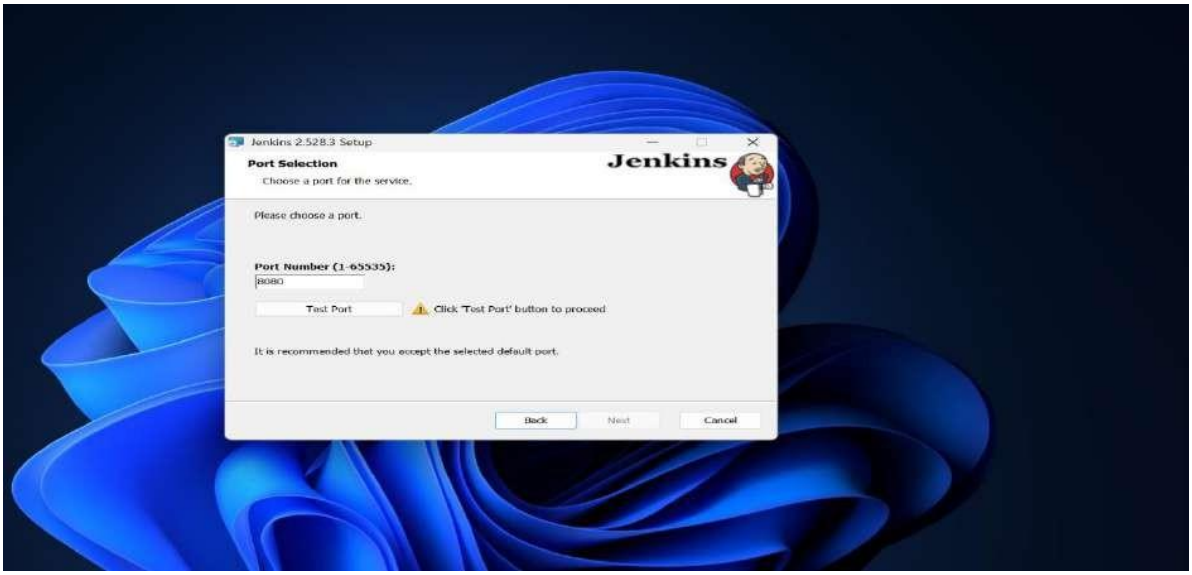
Click on Download



Install .msi file

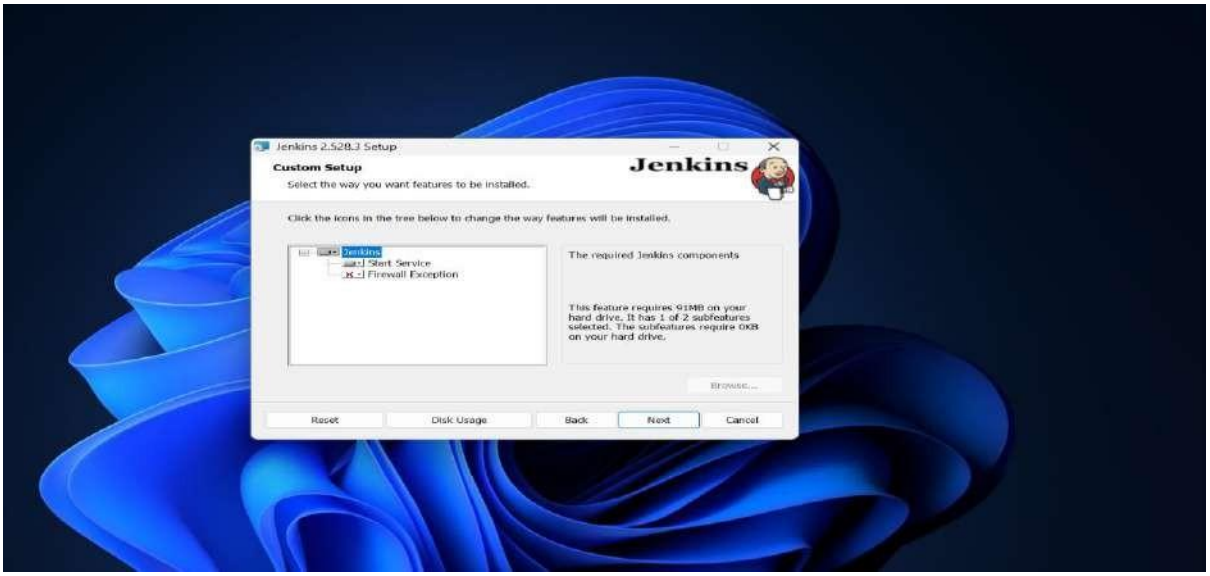

Jenkins 2.528.3 Setup
Destination Folder
Click Next to install to the default folder or click Change to choose another.
Jenkins
Install Jenkins 2.528.3 to:
C:\Program Files\Jenkins\
Change...
Back
Next
Cancel

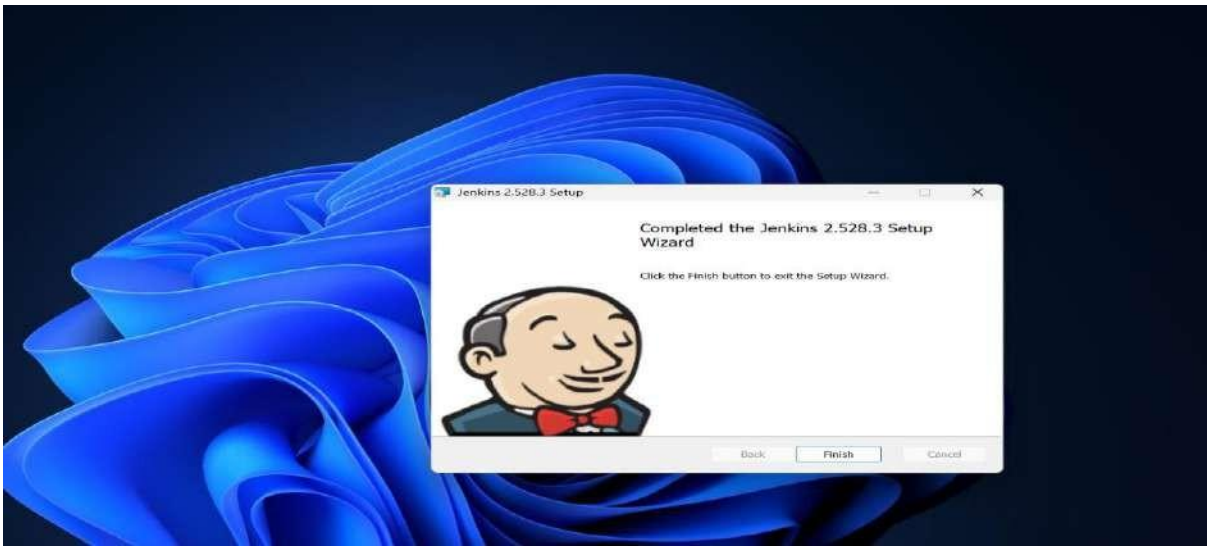
Jenkins 2.528.3 Setup
Service Logon Credentials
Enter service credentials for the service.
Jenkins
Jenkins 2.528.3 installs and runs as an independent Windows service. To operate in this manner, you must supply the user account credentials for Jenkins 2.528.3 to run successfully.
Logon Type:
Run service as LocalSystem (not recommended)
Run service as local or domain user:
Test Credentials
Back
Next
Cancel

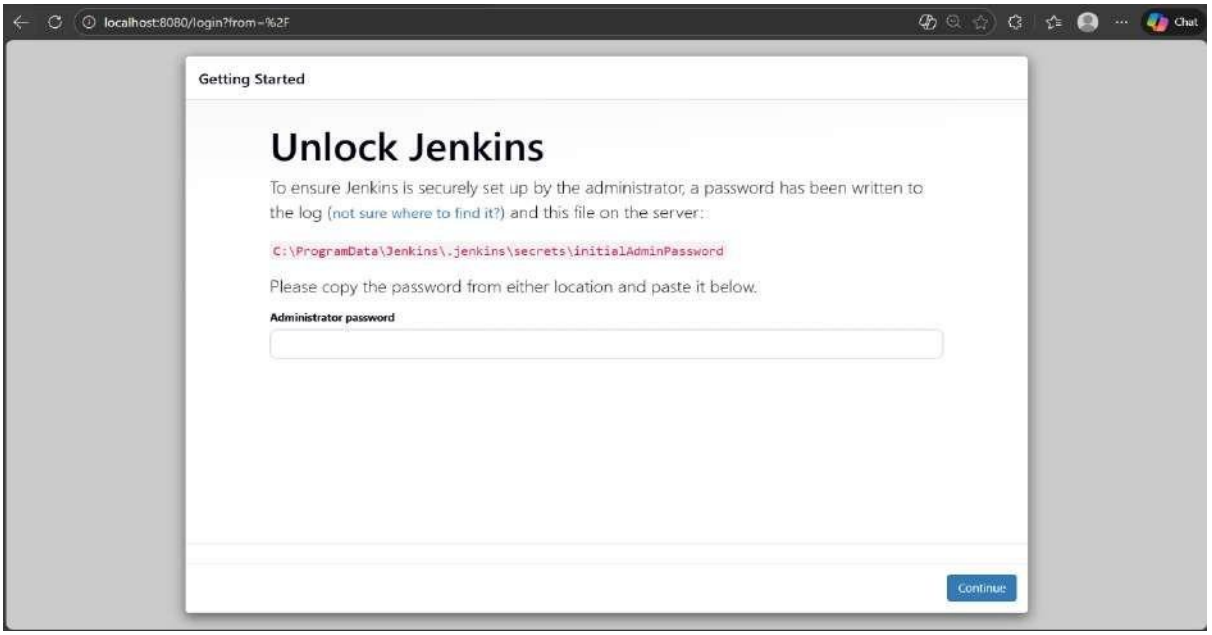

Jenkins 2.528.3 Setup
Port Selection
Choose a port for the service.
Jenkins
Please choose a port.
Port Number (1-65535):
8080
Test Port
Click 'Test Port' button to proceed
It is recommended that you accept the selected default port.
Back
Next
Cancel

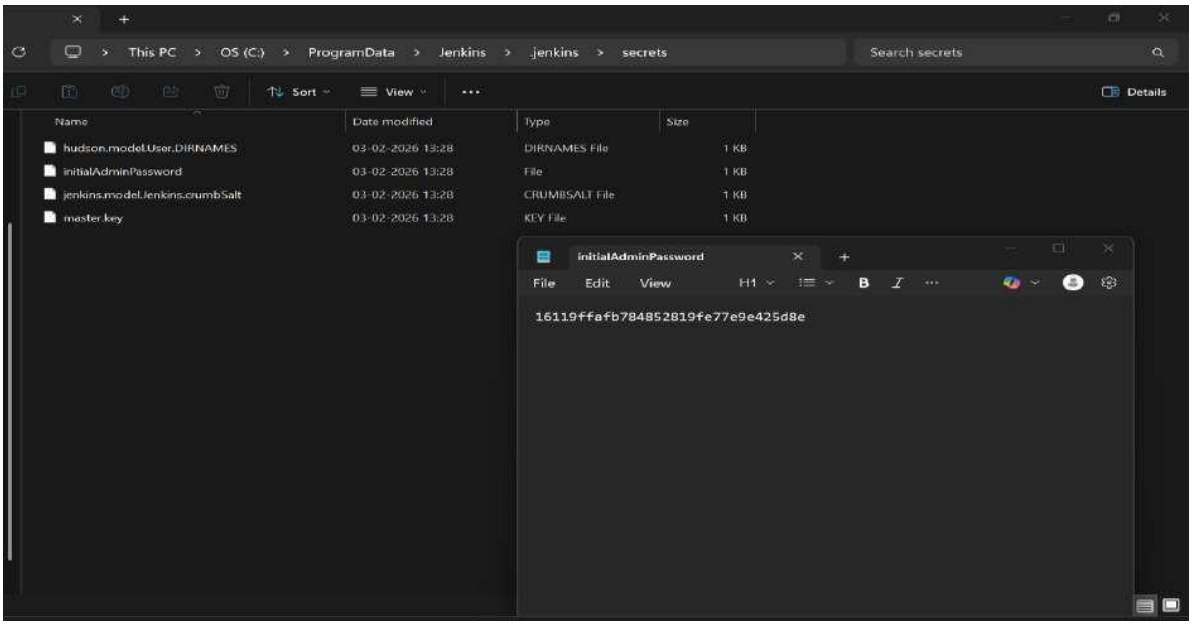
Jenkins 2.528.3 Setup
Port Selection
Choose a port for the service.
Jenkins
Please choose a port.
Port Number (1-65535):
8080
Test Port
It is recommended that you accept the selected default port.
Back
Next
Cancel

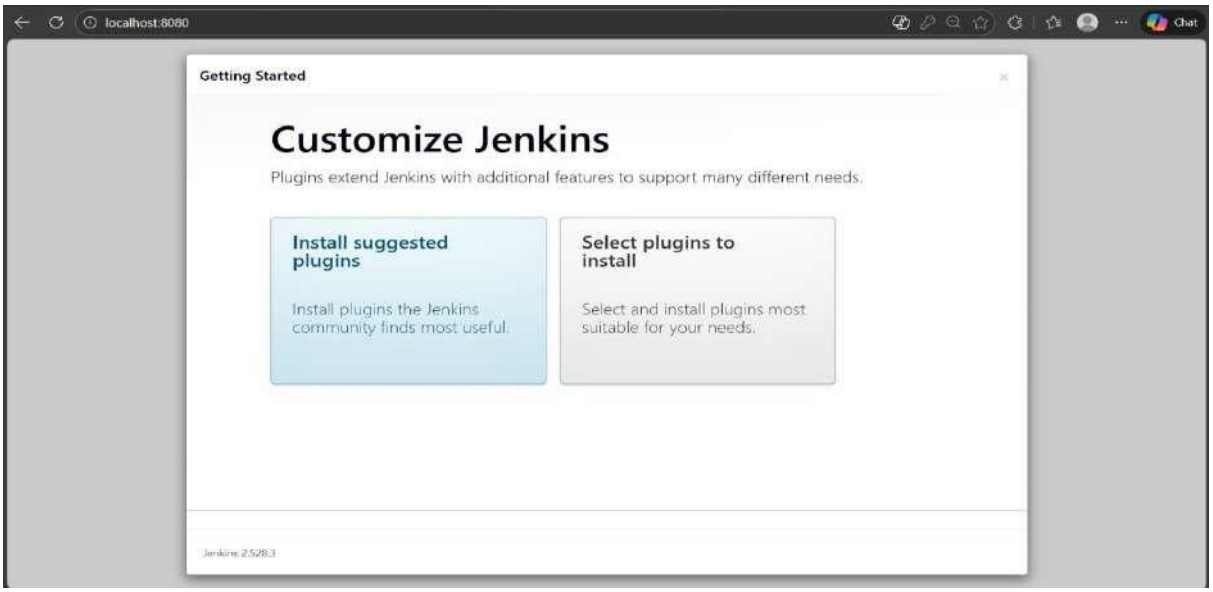
Jenkins 2.528.3 Setup
Jenkins 2.528.3 Setup
Select Java home directory (JDK or JRE)
Jenkins
Please select the path of a Java Development Kit or Java Runtime Environment. Only Java 17 and 21 are supported by Jenkins.
C:\Program Files\Java\jdk-17\
Change...
Back
Next
Cancel

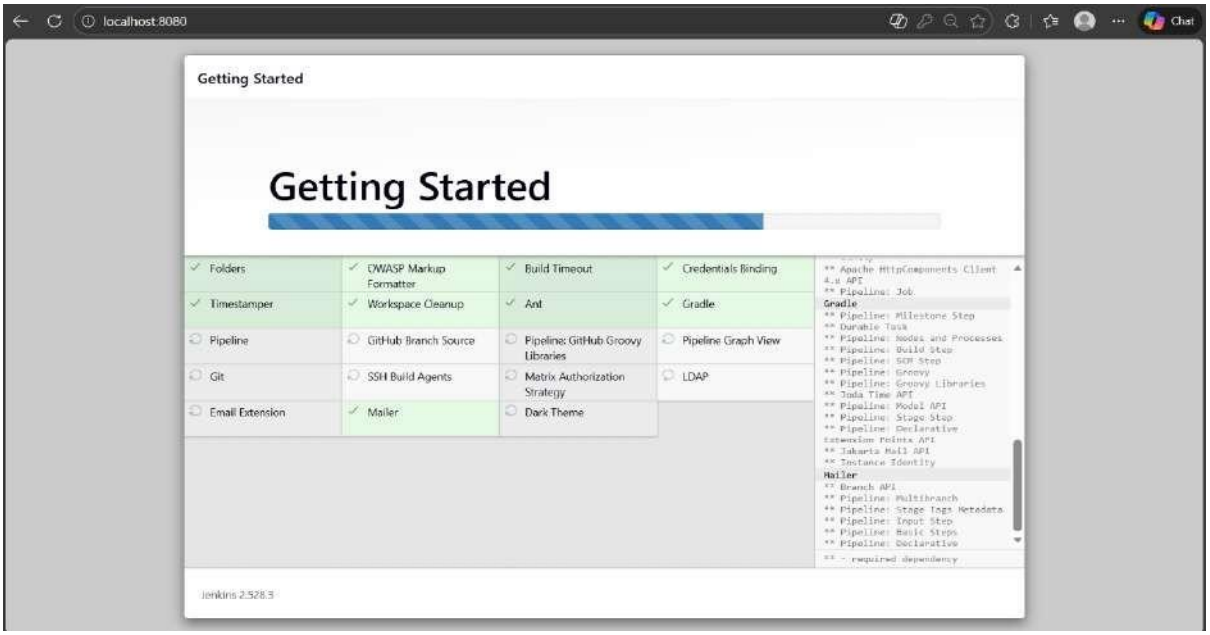

Jenkins 2.528.3 Setup
Custom Setup
Select the way you want features to be installed.
Jenkins
Click the icons in the tree below to change the way features will be installed.
Jenkins
Start Service
Firewall Exception
The required Jenkins components
This feature requires 91MB on your hard drive. It has 1 of 2 subfeatures selected. The subfeatures require 0KB on your hard drive.
Browse...
Reset
Disk Usage
Back
Next
Cancel

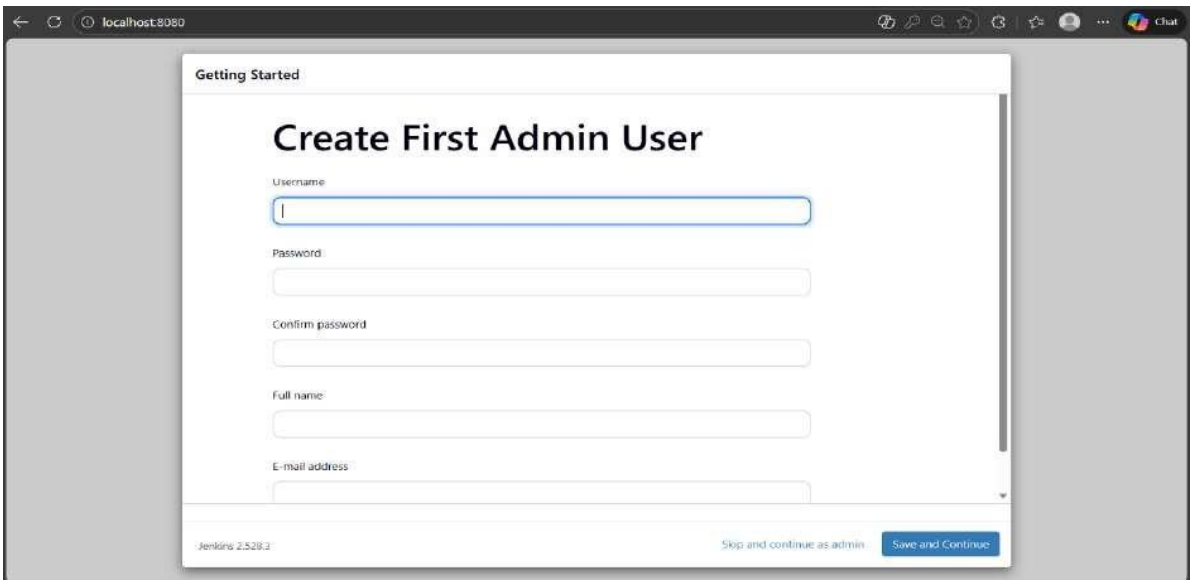

Jenkins 2.528.3 Setup
Completed the Jenkins 2.528.3 Setup Wizard
Click the Finish button to exit the Setup Wizard.
Back
Finish
Cancel

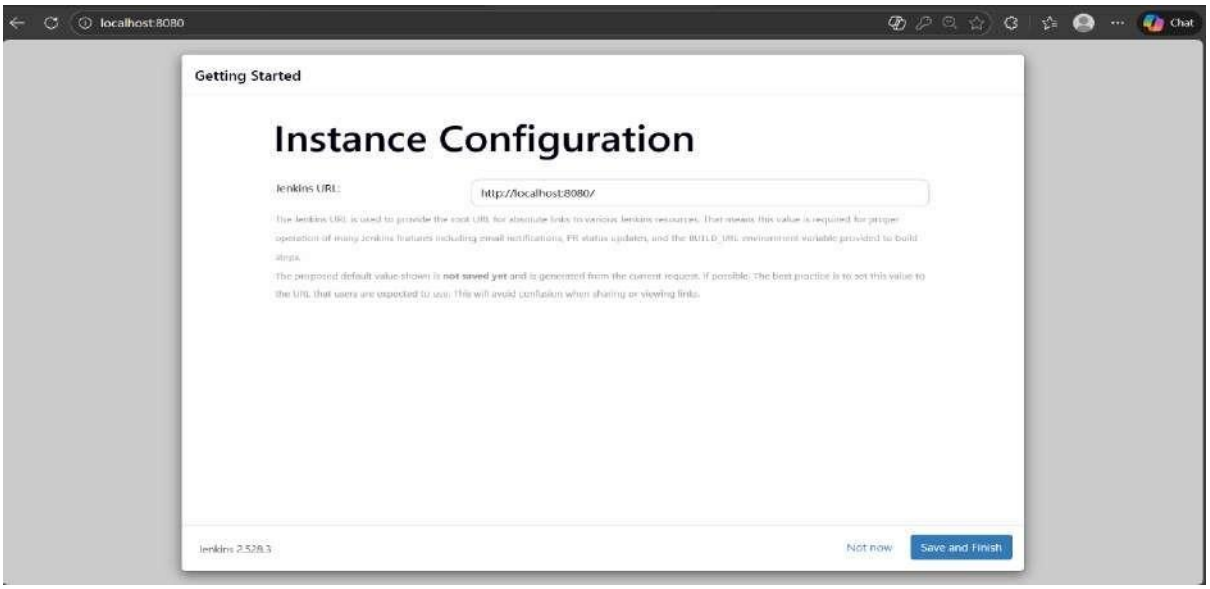

localhost:8080/login?from=%2F
Chat
Getting Started
Unlock Jenkins
To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:
C:\ProgramData\Jenkins\.jenkins\secrets\initialAdminPassword
Please copy the password from either location and paste it below.
Administrator password
Continue

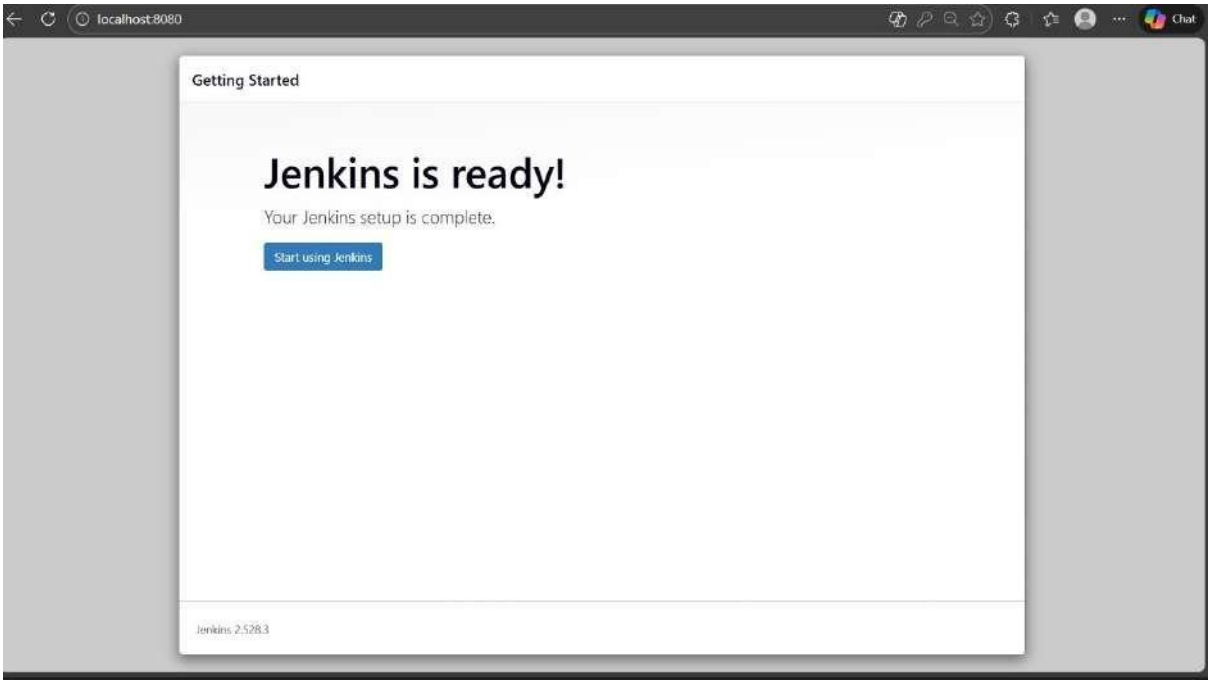

This PC > OS (C:) > ProgramData > Jenkins > .jenkins > secrets
Search secrets
Sort
View
Details
Name
Date modified
Type
Size
hudson.model.User.DIRNAMES
03-02-2026 13:28
DIRNAMES File
1 KB
initialAdminPassword
03-02-2026 13:28
File
1 KB
jenkins.model.Jenkins.crumbSalt
03-02-2026 13:28
CRUMBSALT File
1 KB
master.key
03-02-2026 13:28
KEY File
1 KB
initialAdminPassword
File
Edit
View
H1
16119ffafb784852819fe77e9e425d8e


localhost:8080
Getting Started
Customize Jenkins
Plugins extend Jenkins with additional features to support many different needs.
Install suggested plugins
Install plugins the Jenkins community finds most useful.
Select plugins to install
Select and install plugins most suitable for your needs.
Jenkins 2.528.3


localhost:8080
Getting Started
Getting Started
Folders
OWASP Markup Formatter
Build Timeout
Credentials Binding
Timestamper
Workspace Cleanup
Ant
Gradle
Pipeline
GitHub Branch Source
Pipeline: GitHub Groovy Libraries
Pipeline Graph View
Git
SSH Build Agents
Matrix Authorization Strategy
LDAP
Email Extension
Mailer
Dark Theme
Jenkins 2.528.3

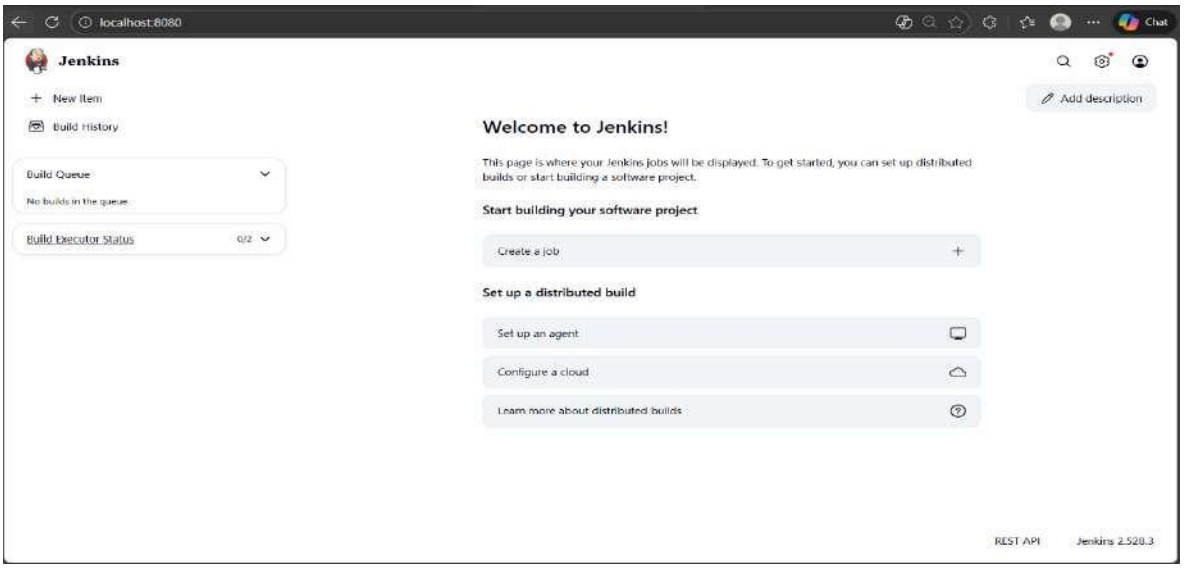

localhost:8080
Getting Started
Create First Admin User
Username
Password
Confirm password
Full name
E-mail address
Jenkins 2.528.3
Skip and continue as admin
Save and Continue



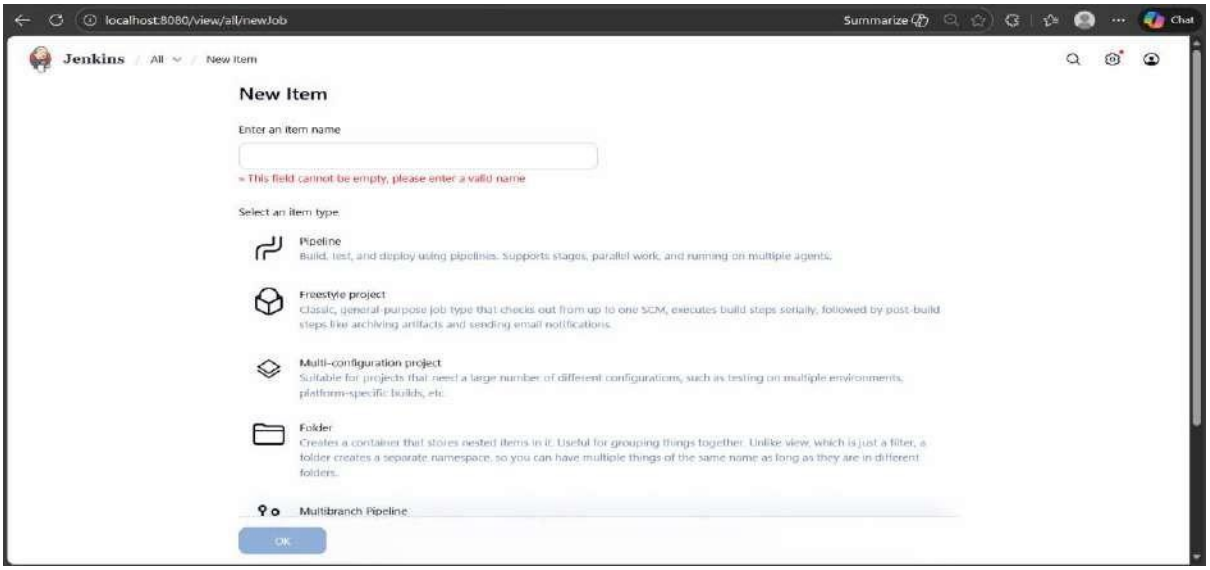


3. Jenkins UI overview

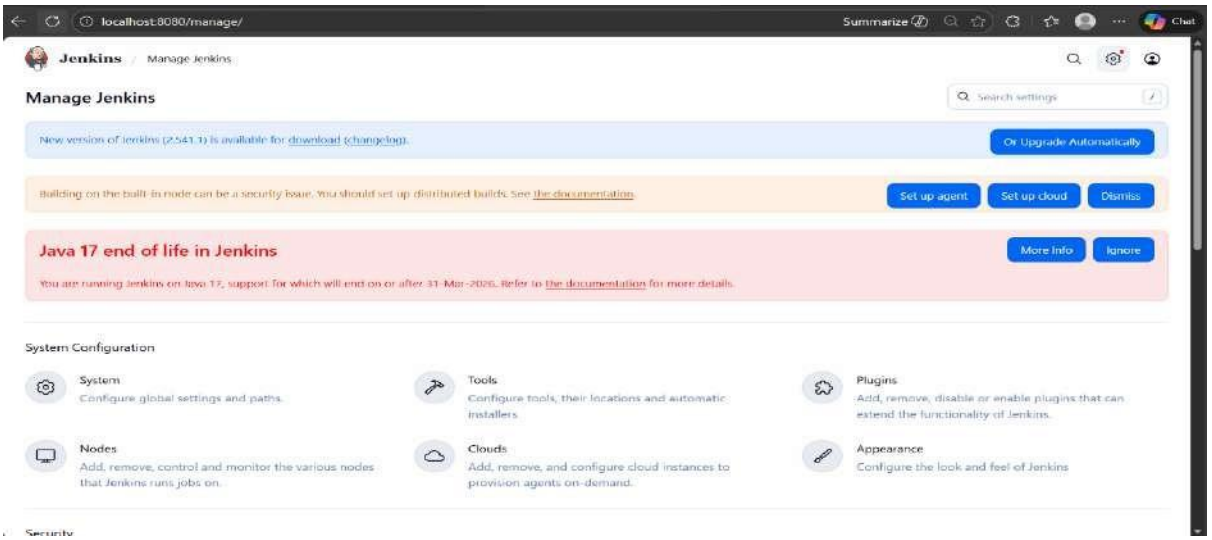
Dashboard: Shows all jobs and their status.



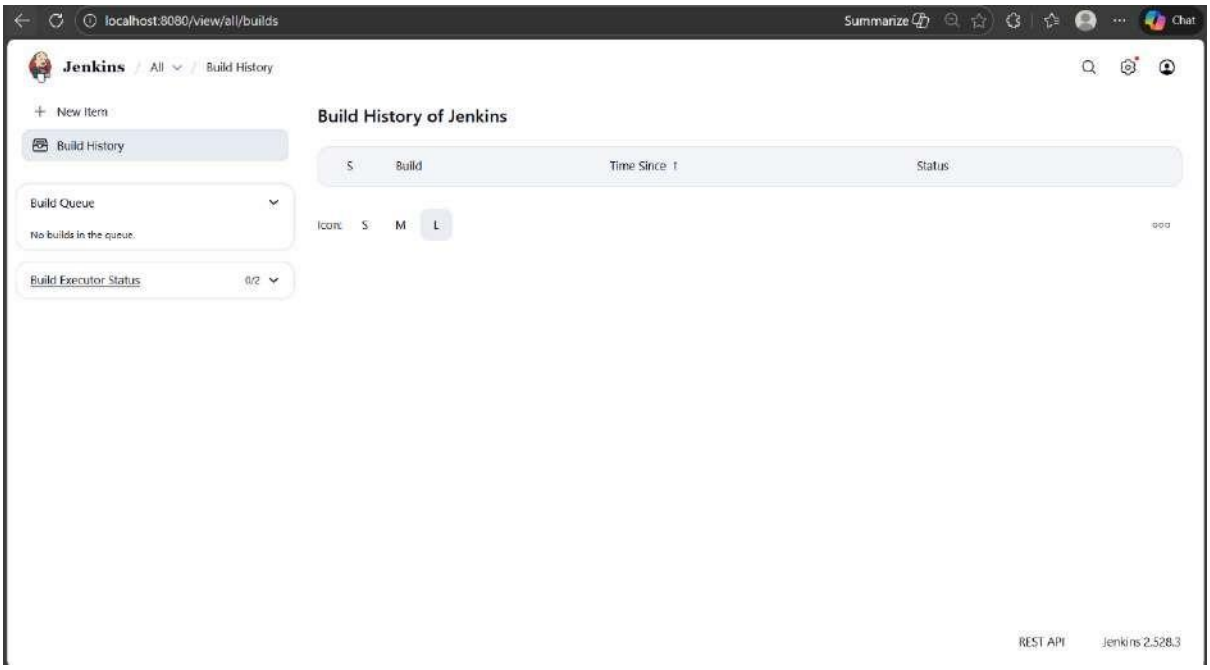
New Item: Create a new job.



Manage Jenkins: Configure Jenkins, plugins, and system settings.

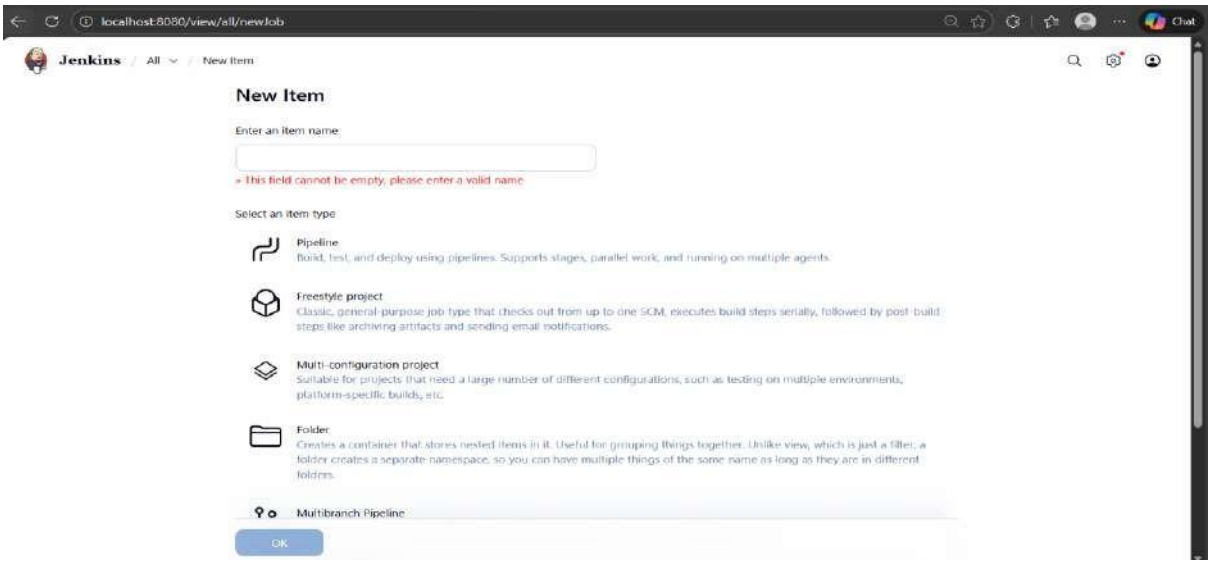


Build Queue / Build History: Shows running and past jobs.

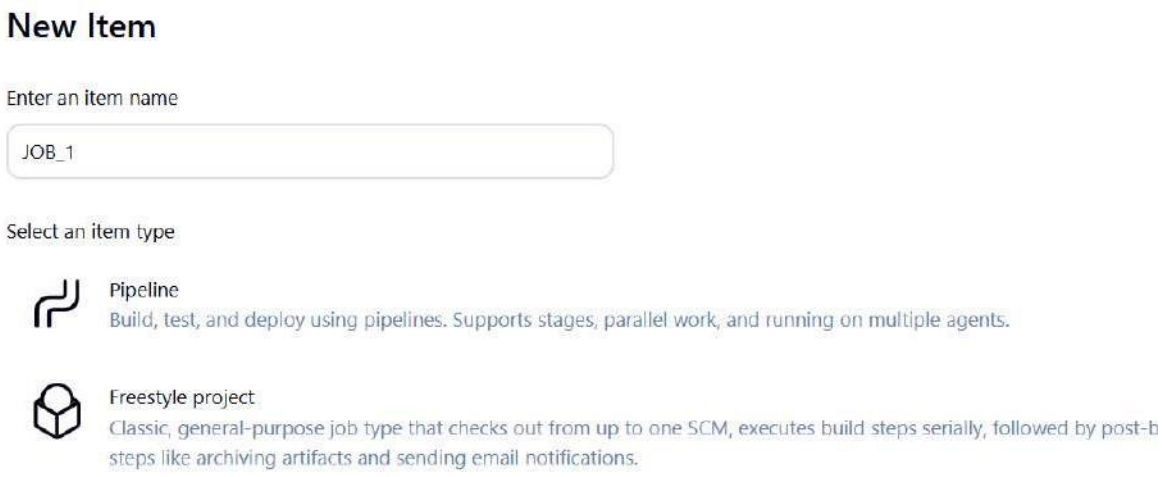


4. **Create Freestyle Job**

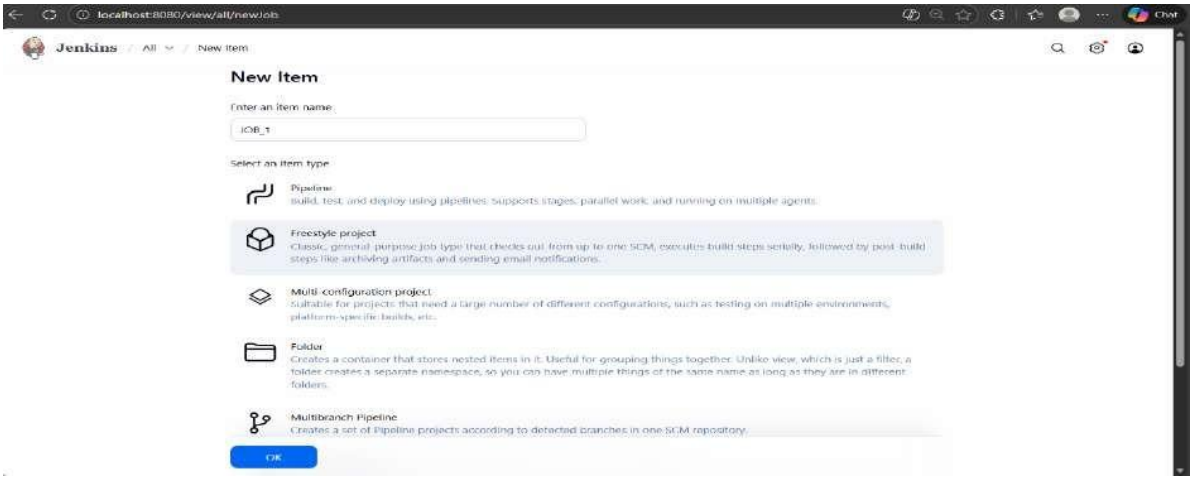
   Click **"New Item"** on dashboard.



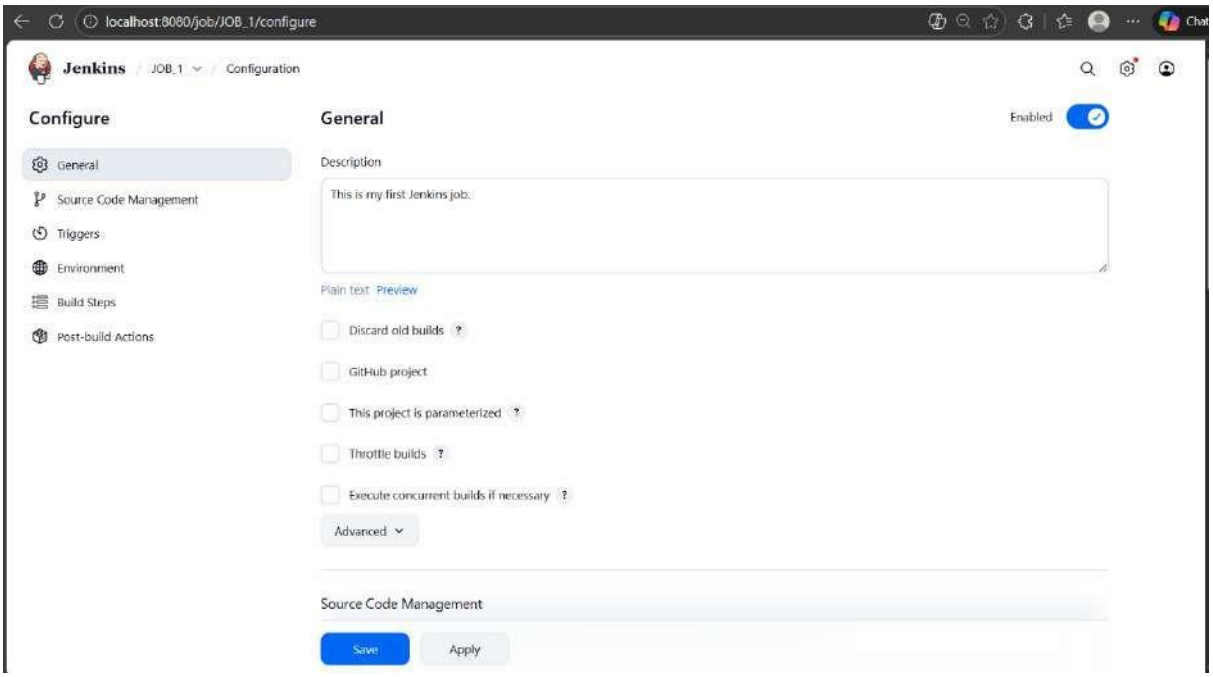
Enter a name.



Select **Freestyle project** → click **OK**.

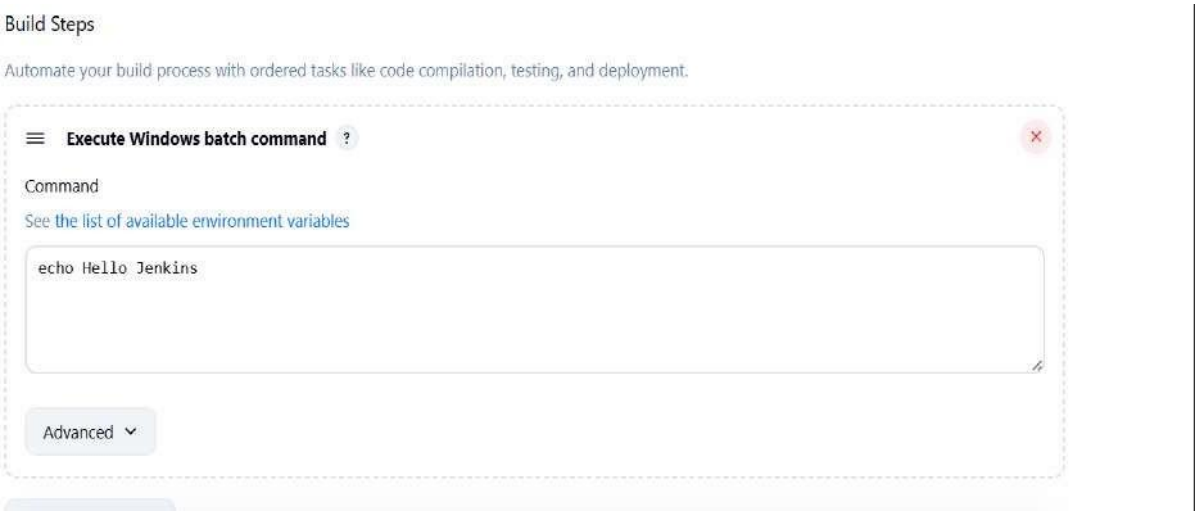


Configure description if needed.

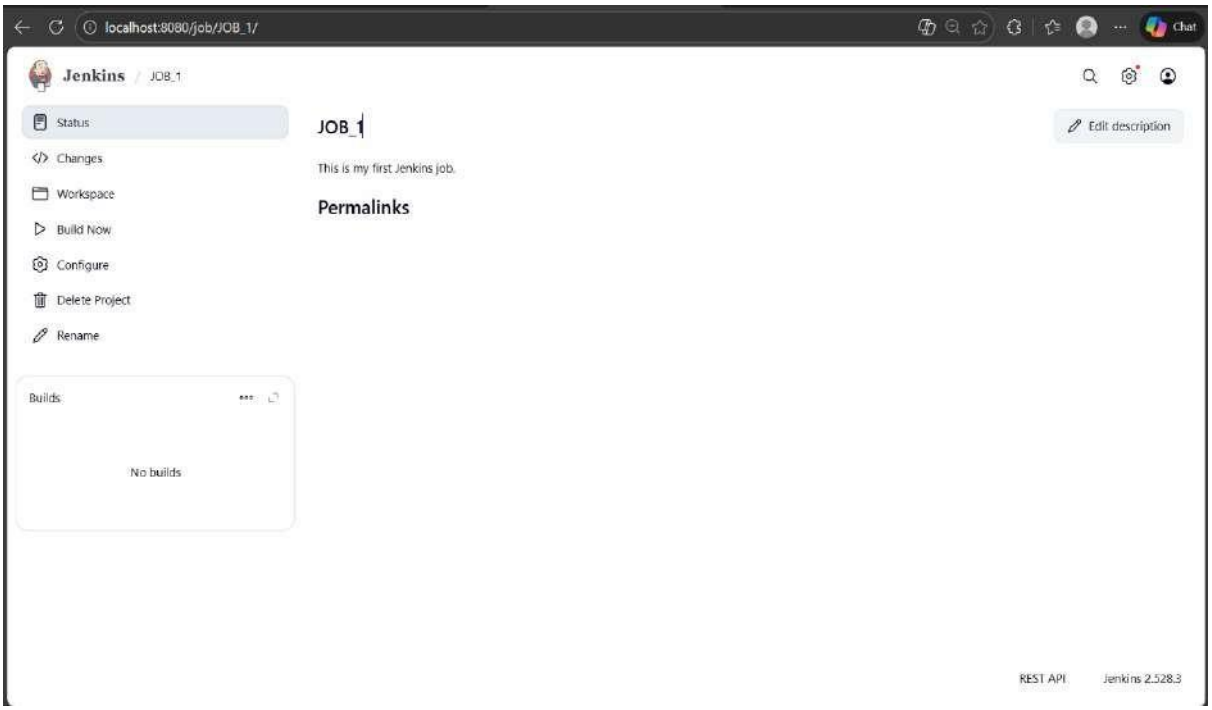


5. **Add simple build step:**

Windows: echo Hello Jenkins

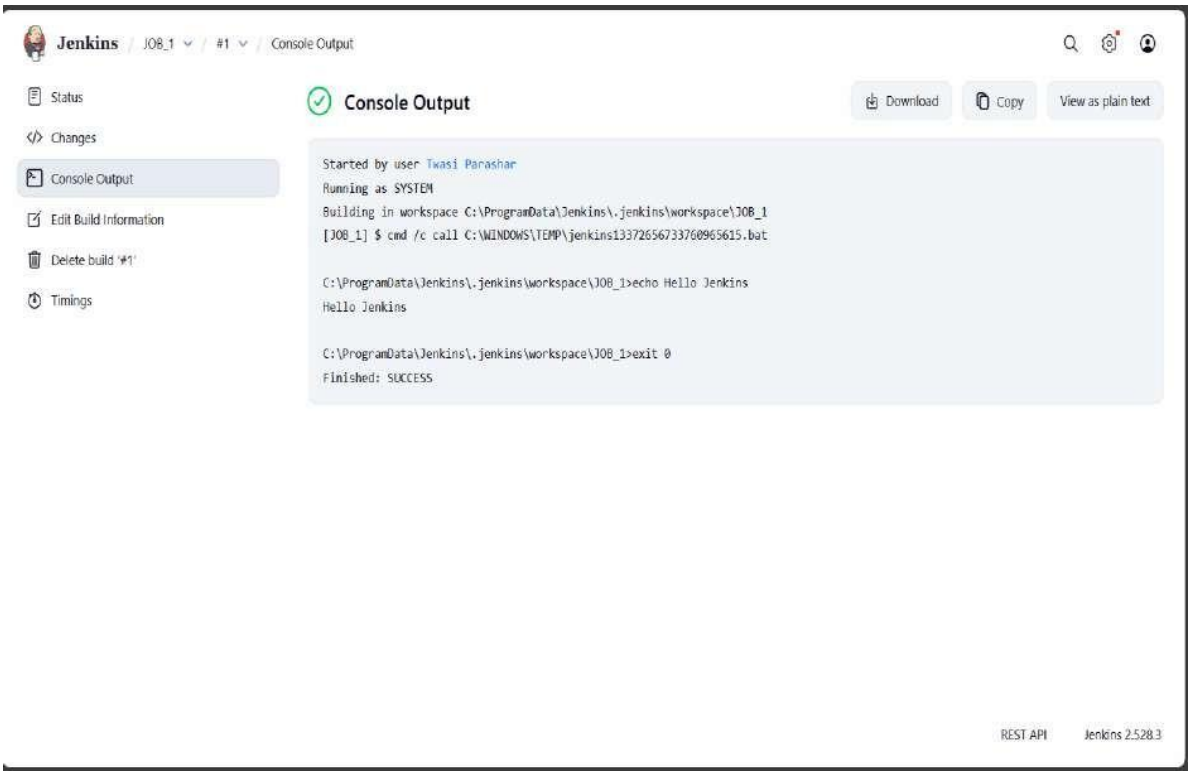


6. **Run job manually**



7. **View:**

Console Output

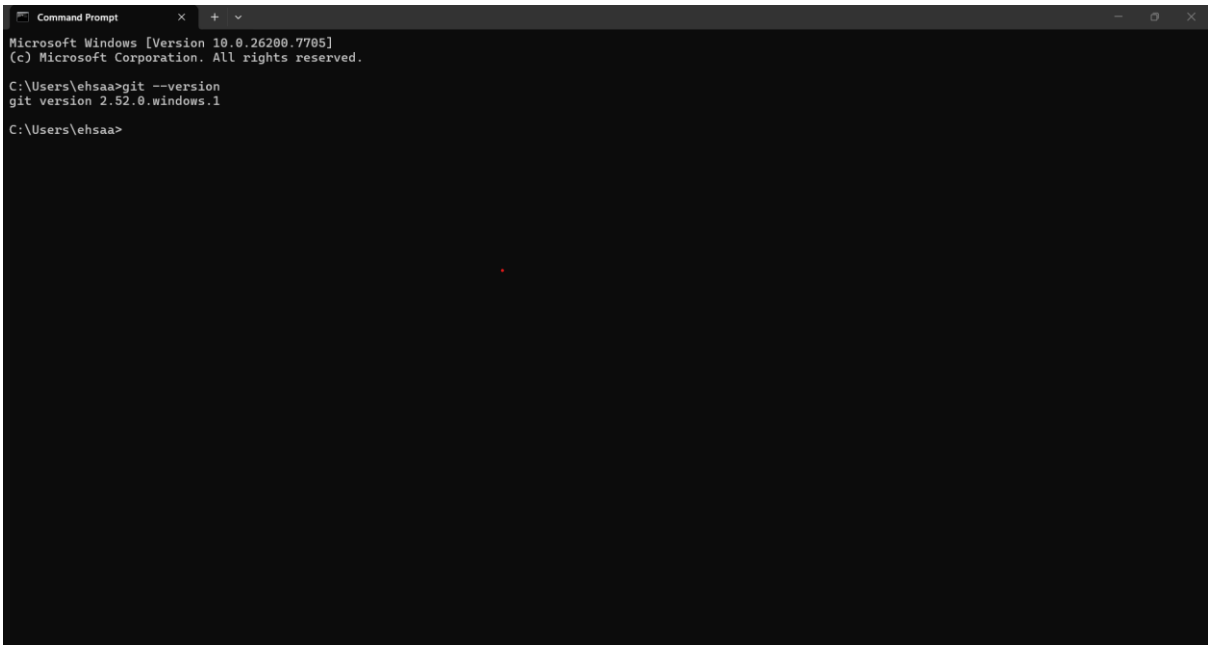


Build History

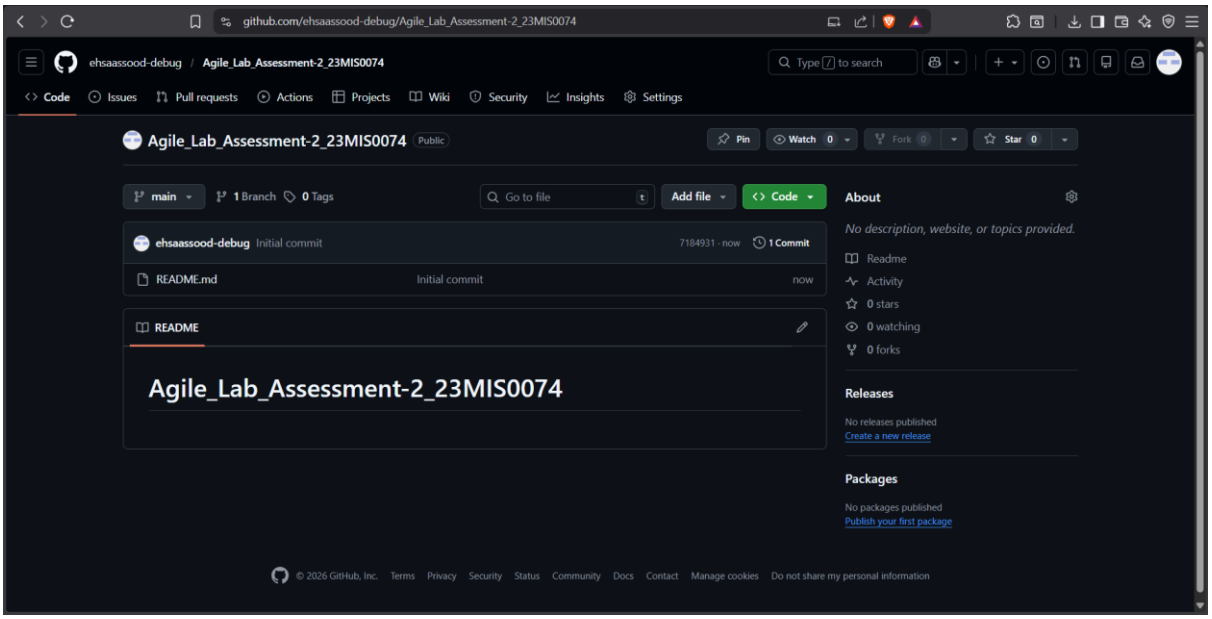
2. **Learning Goal:** Work with source code and build automation

   Stepwise tasks

   1. **Install Git**

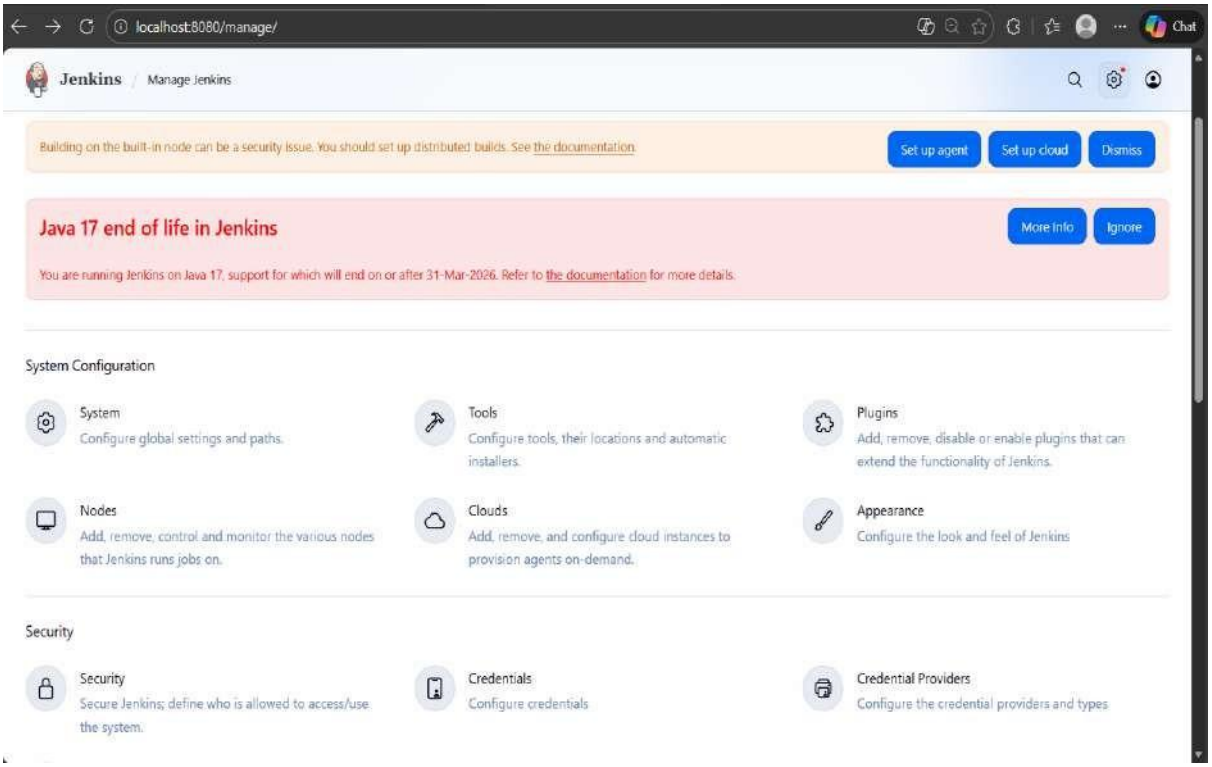


2. Create GitHub repository

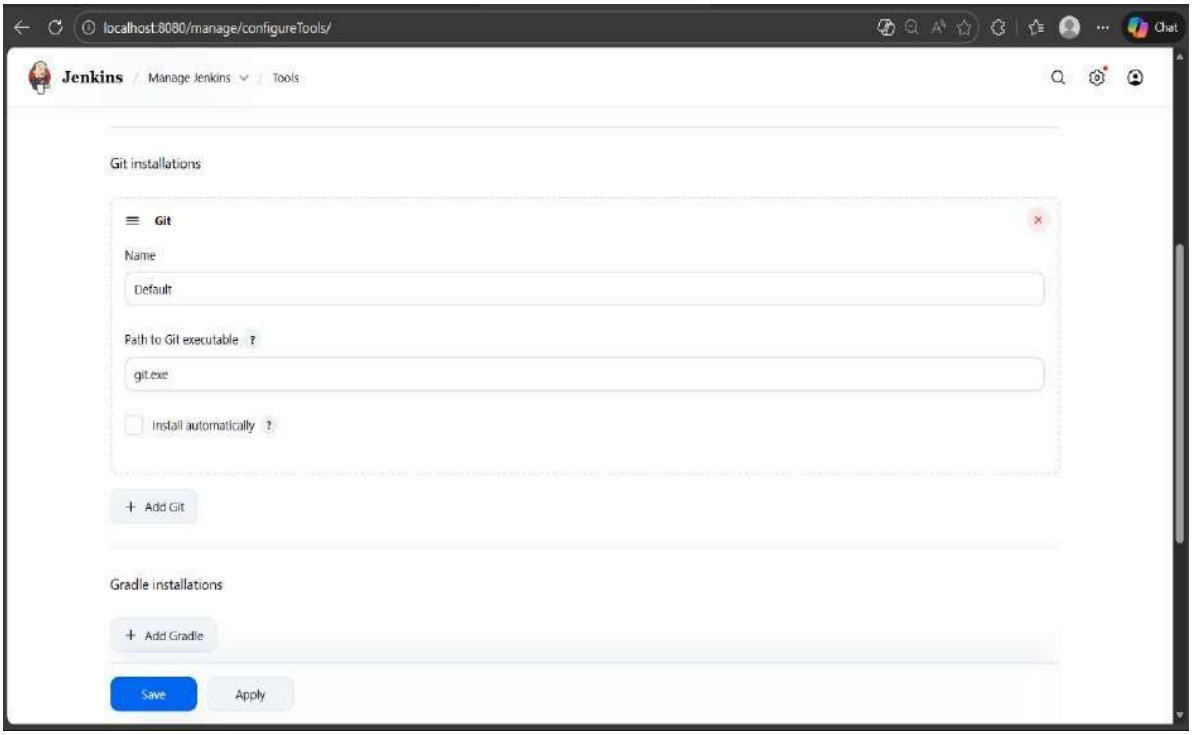


3. Configure Git in Jenkins

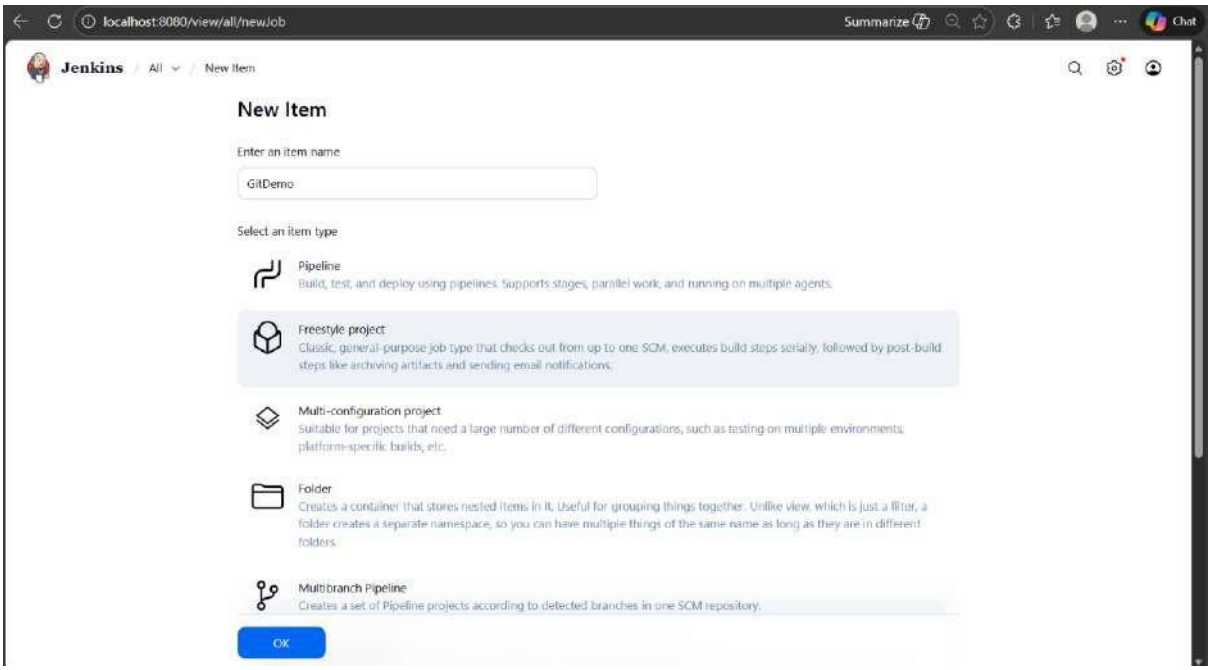
Go to Manage Jenkins → Global Tool Configuration.



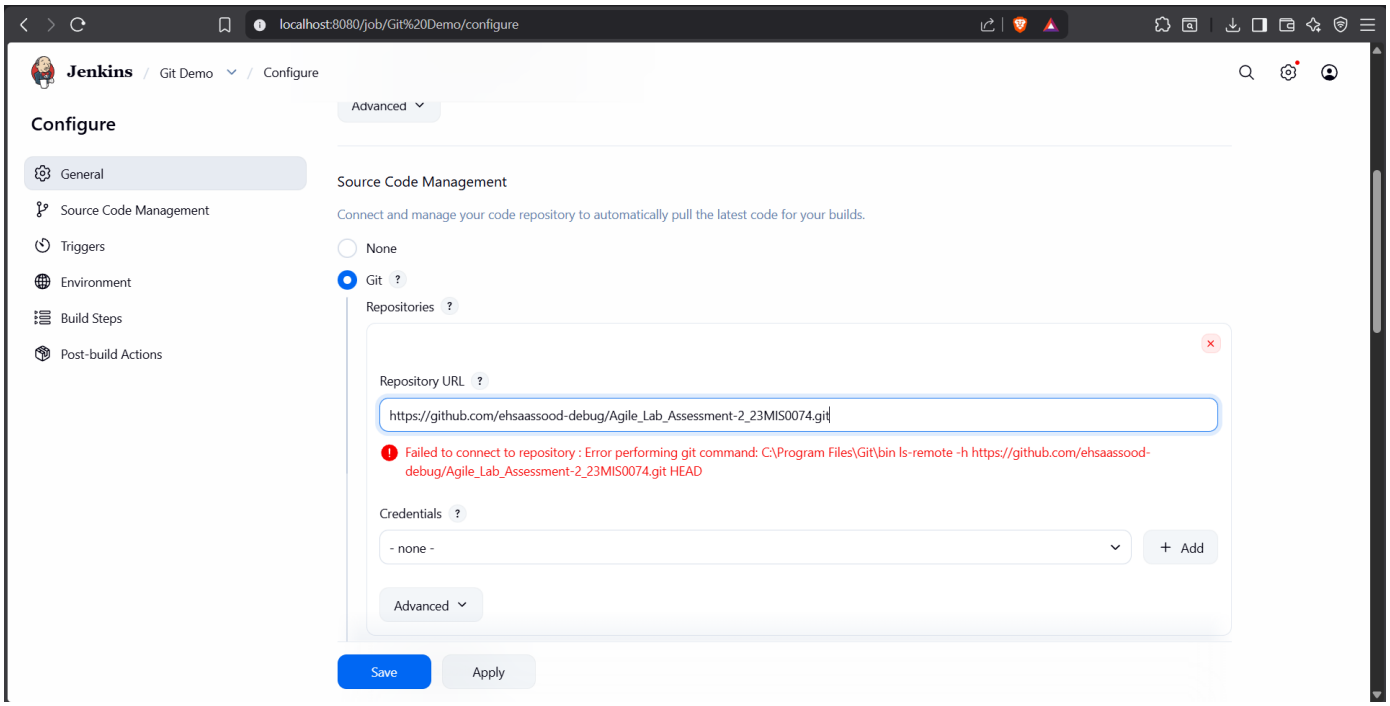
Under Git, click Add Git → provide:



4. Create Freestyle Job:

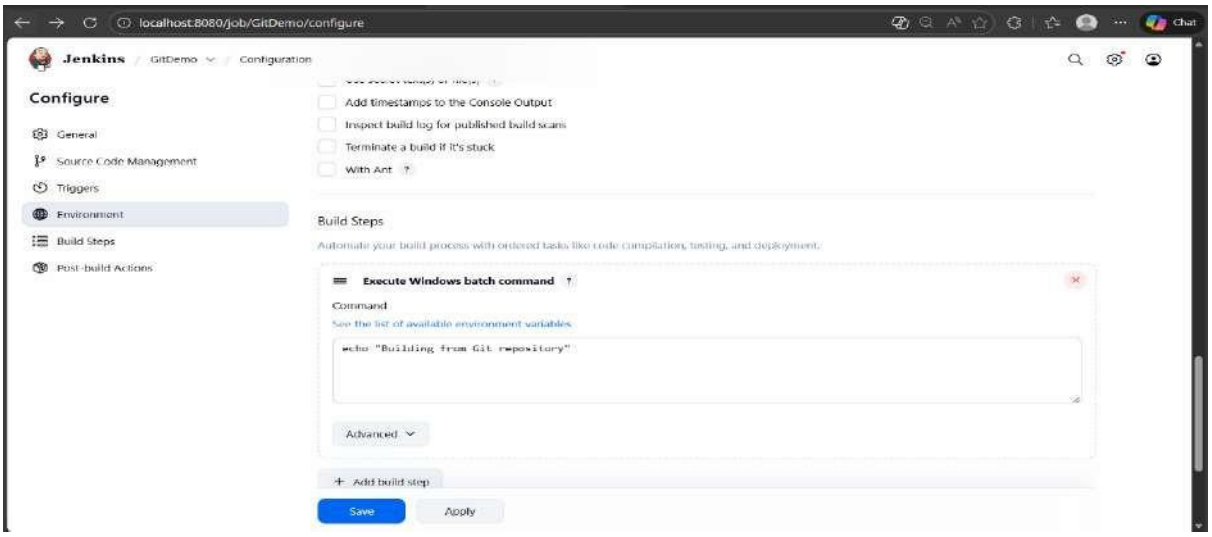


Source Code Management → Git

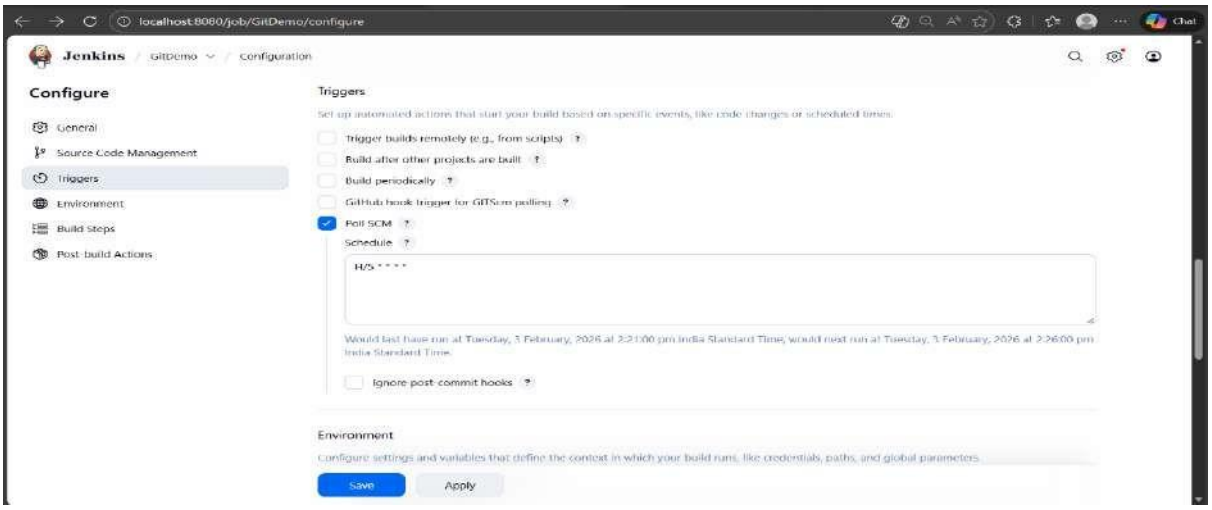


5. Add build step:

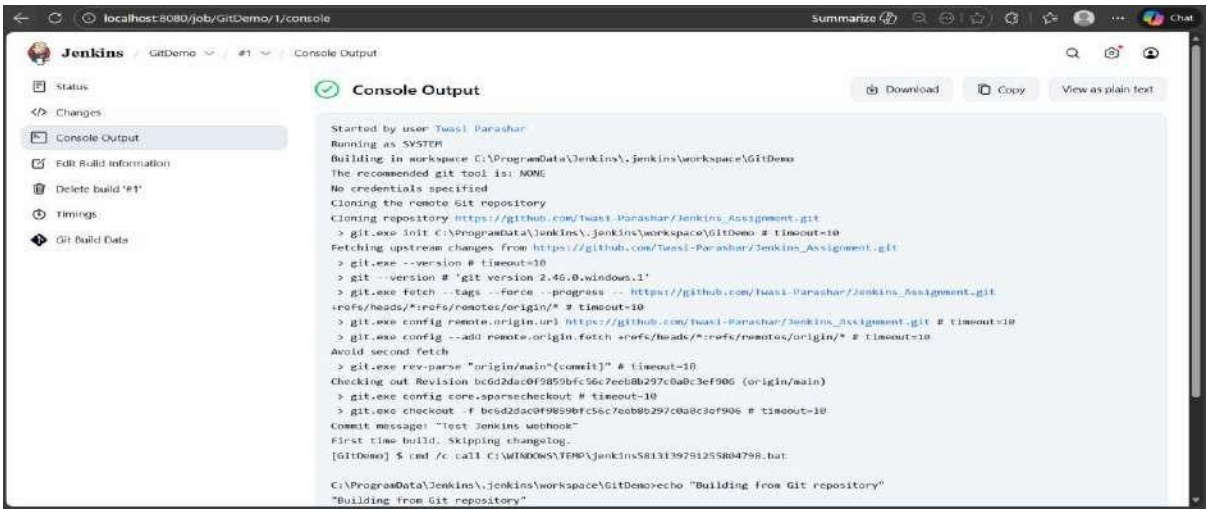
Windows batch / Shell



6. Poll SCM / Webhook trigger



7. Build automatically on code change

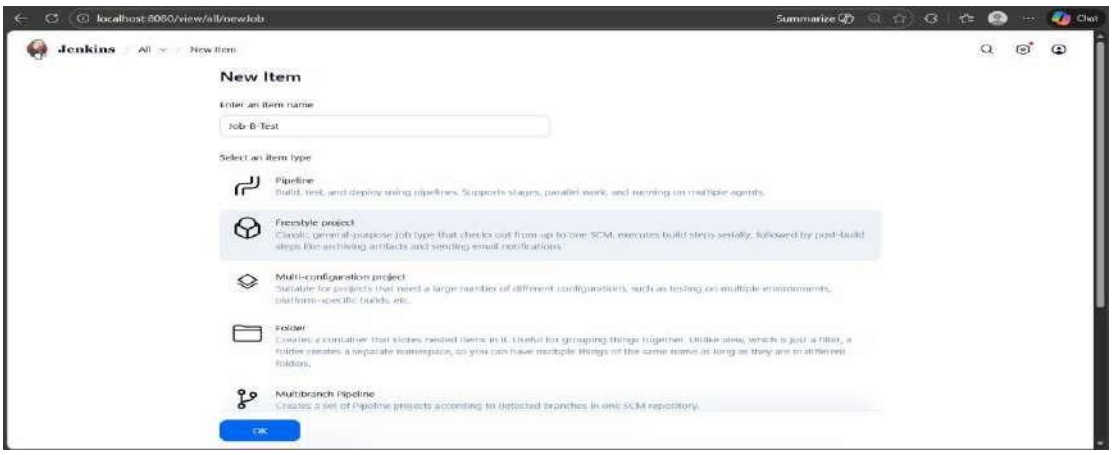


3. **Learning Goal:** Understand job flow visually

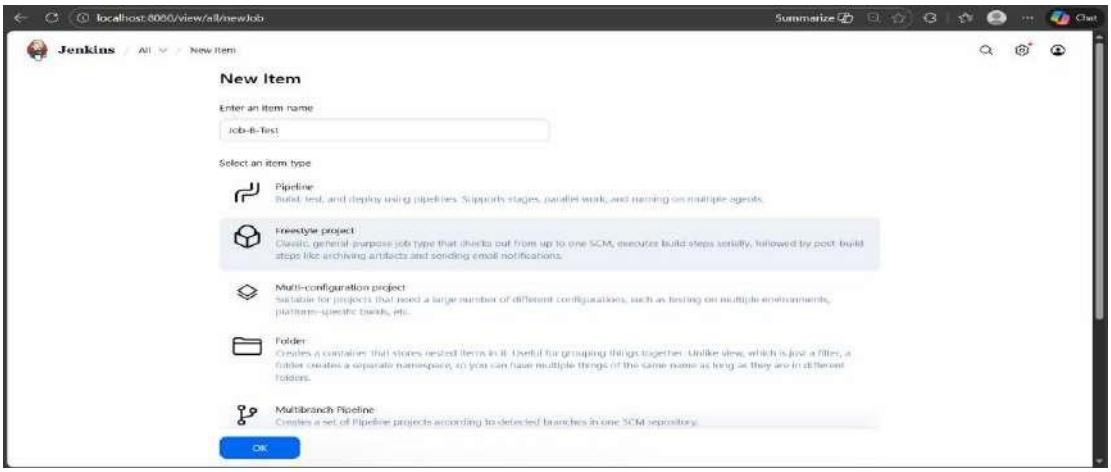
Stepwise Tasks

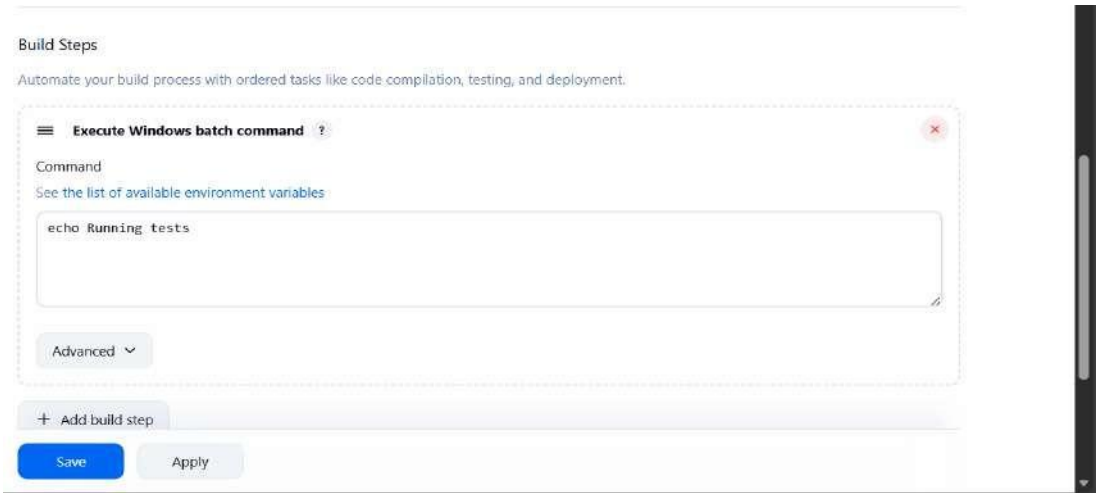
1. **Create multiple Freestyle jobs:**

Job-A (Build)

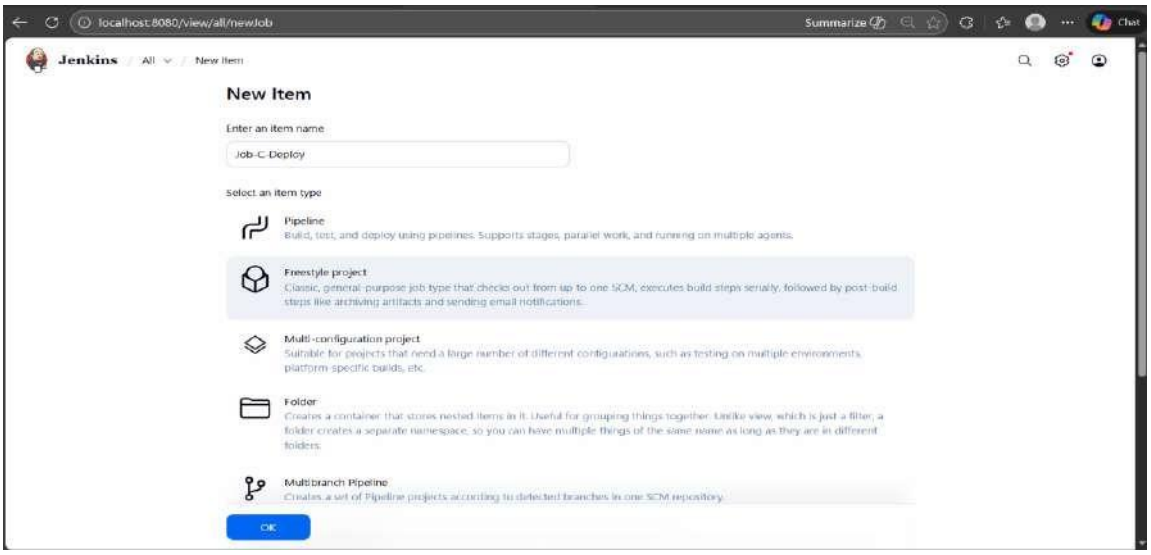


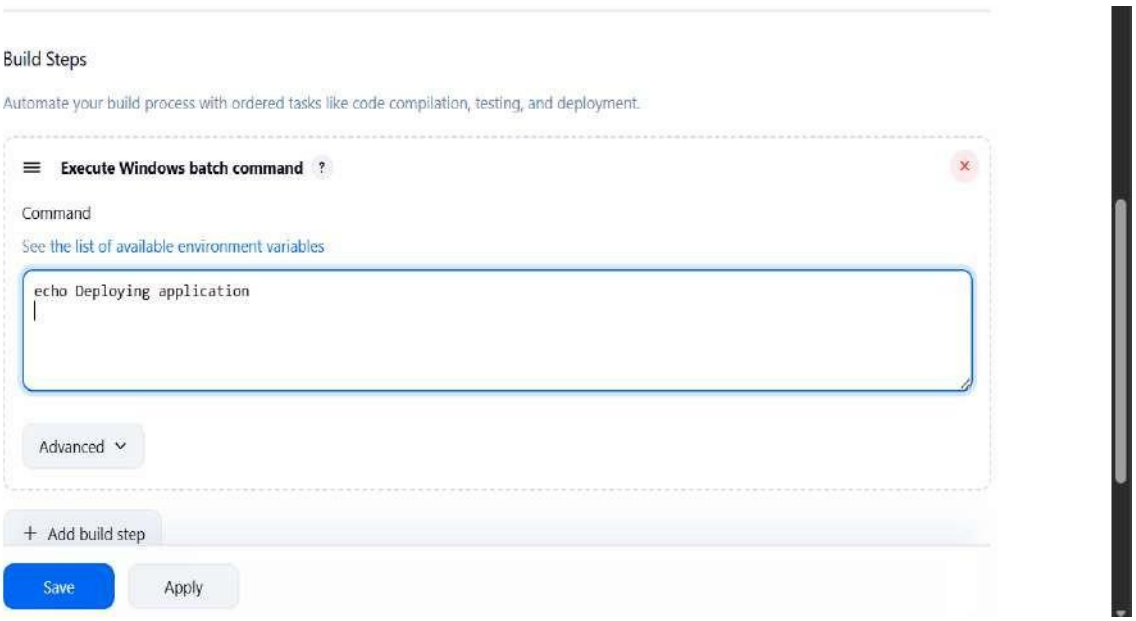
Job-B (Test)




Build Steps
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Execute Windows batch command
Command
See the list of available environment variables
echo Running tests
Advanced
Add build step
Save
Apply

Job-C (Deploy)


localhost:8080/view/all/newJob
Jenkins
New Item
Enter an item name
Job-C Deploy
Select an item type
Pipeline
Freestyle project
Multi-configuration project
Folder
Multibranch Pipeline
OK

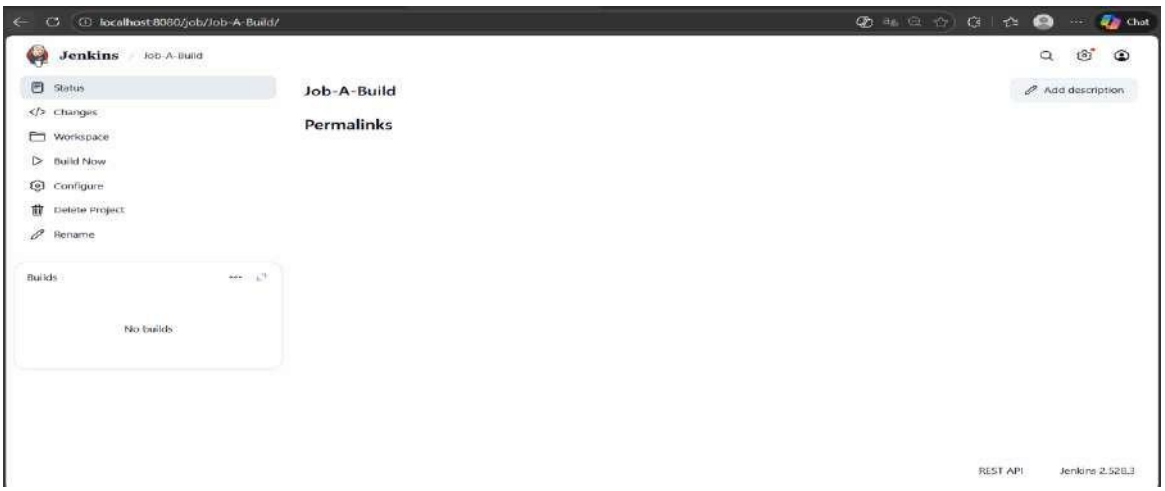

Build Steps
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Execute Windows batch command
Command
See the list of available environment variables
echo Deploying application
Advanced
Add build step
Save
Apply

2. **Use:**

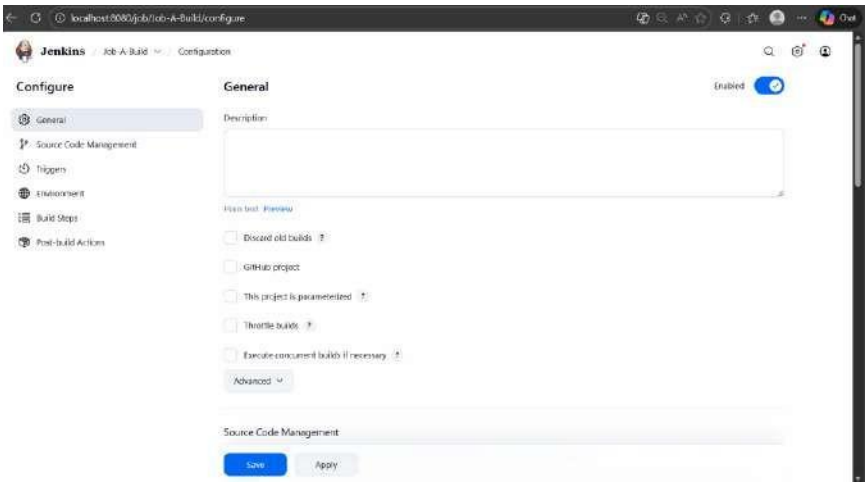
Post-build action → Trigger other projects

Configure Job-A → Trigger Job-B

Open Job-A-Build

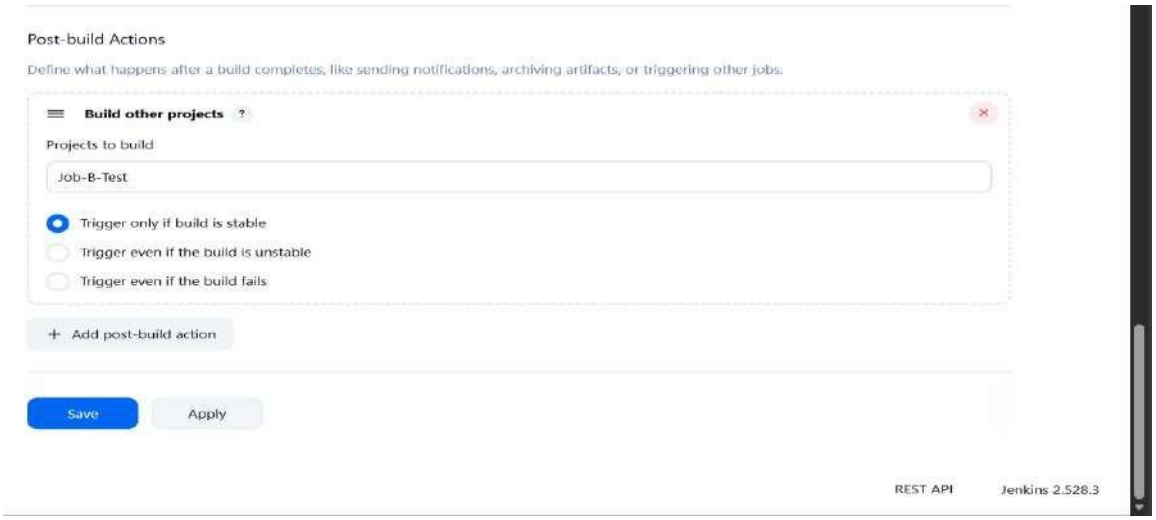


Click Configure



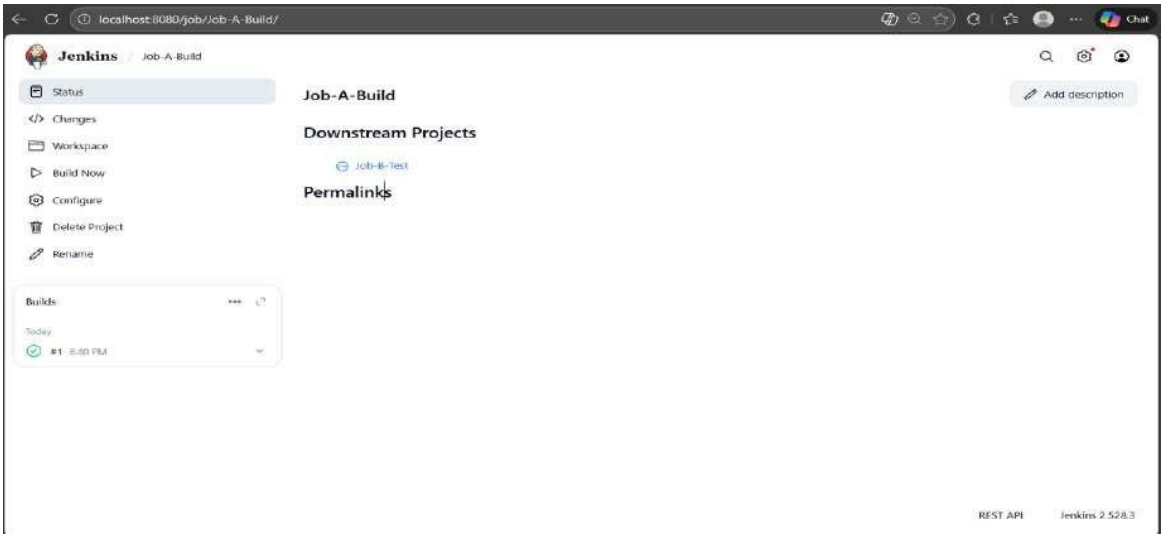
Scroll to Post-build Actions, Click Add post-build action and Select Build other projects

Project to build and Trigger only if build is stable (default)



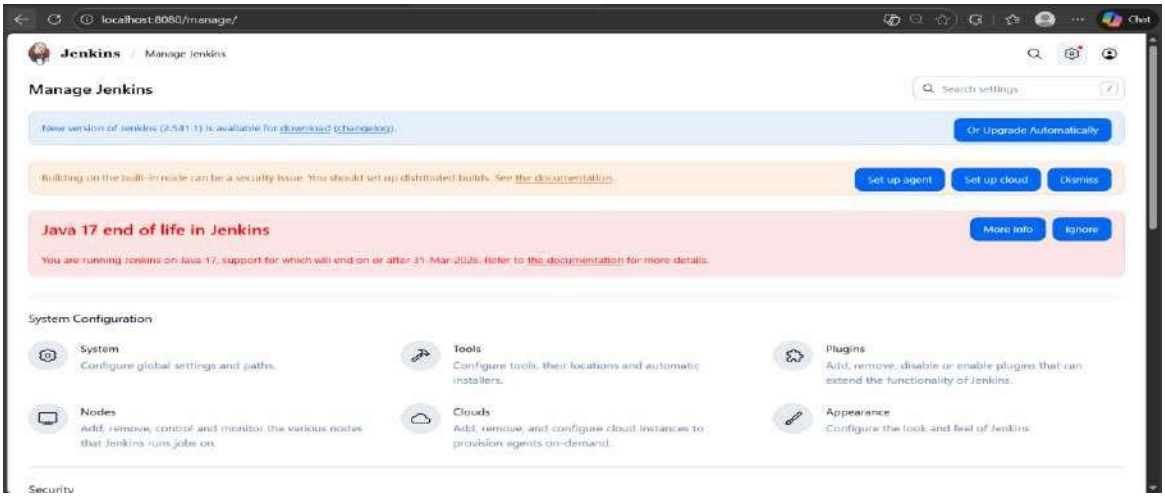
Configure Job-B → Trigger Job-C

3. View job chaining visually

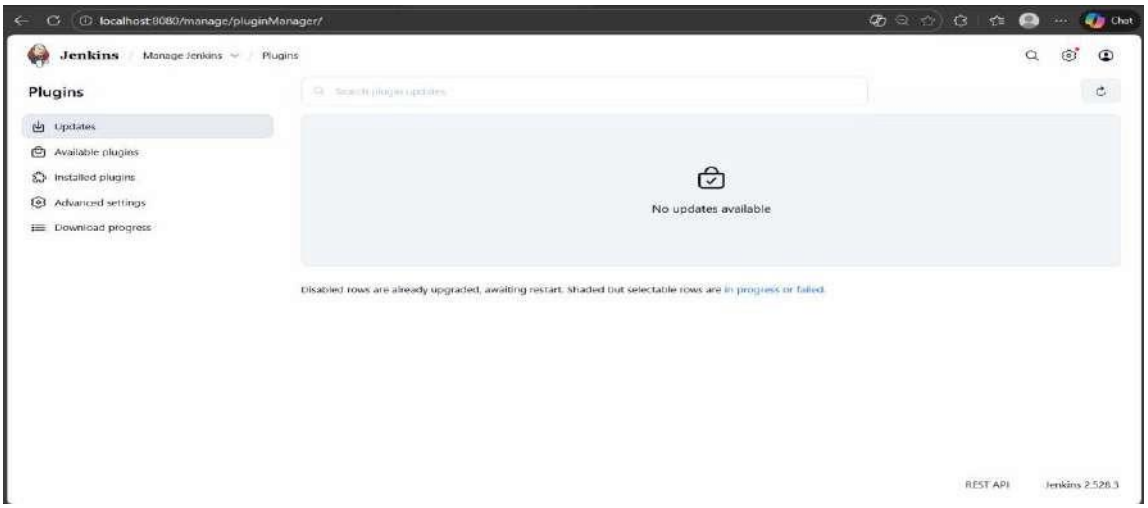


4. Install Pipeline Stage View

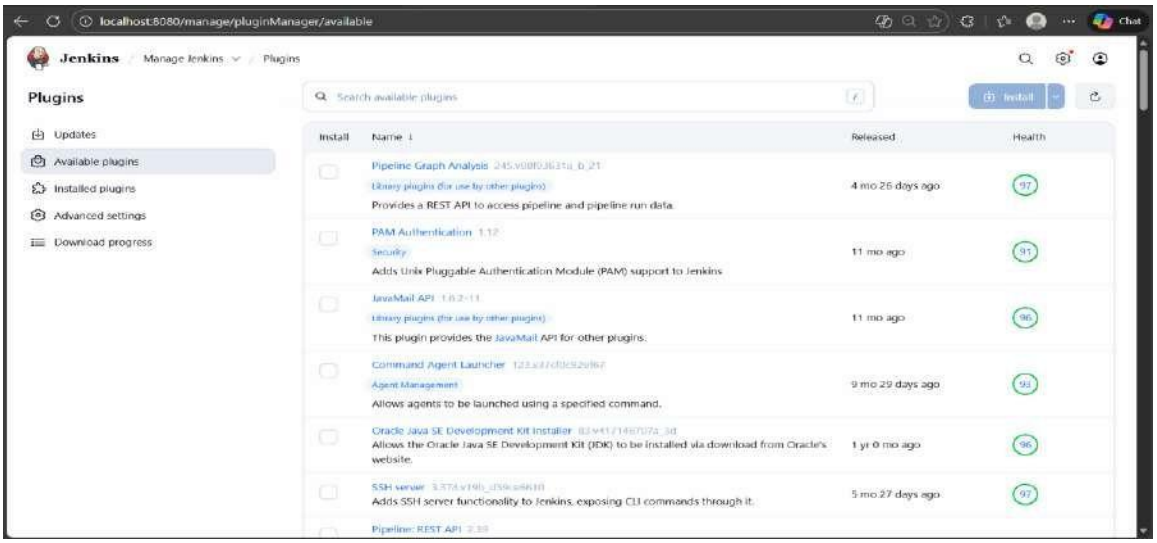
Go to Manage Jenkins



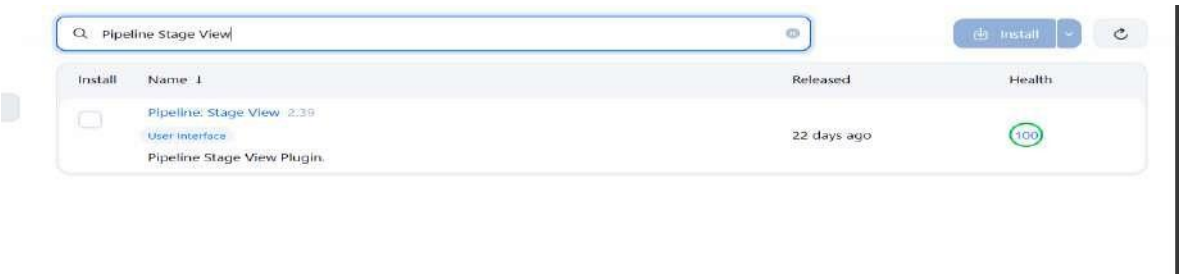
Click Manage Plugins



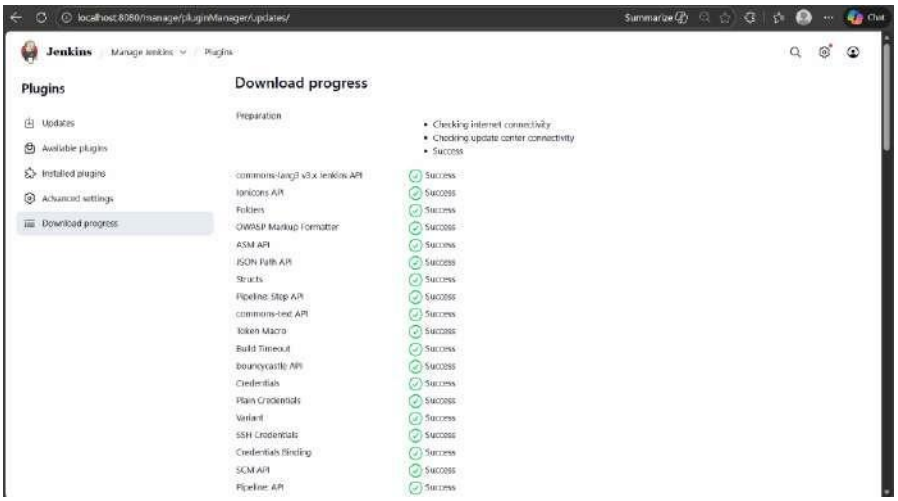
Go to Available



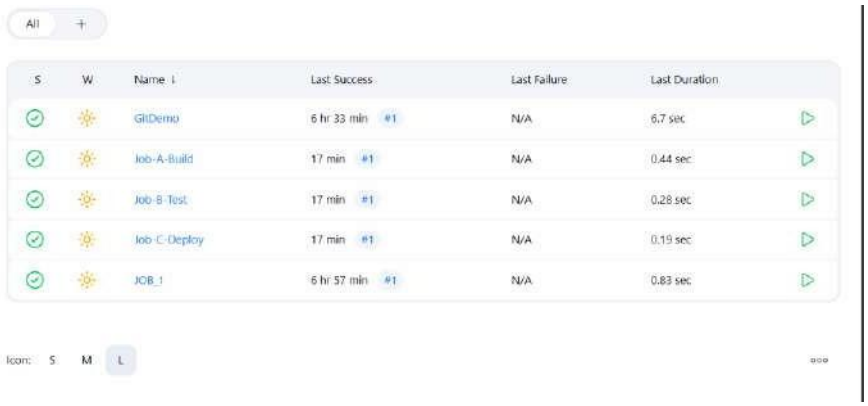
Search: Pipeline Stage View



Select it → Install without restart



5. Observe stage-wise execution



4. **Learning Goal:** Use Jenkins Pipeline & Jenkinsfile
Stepwise Tasks

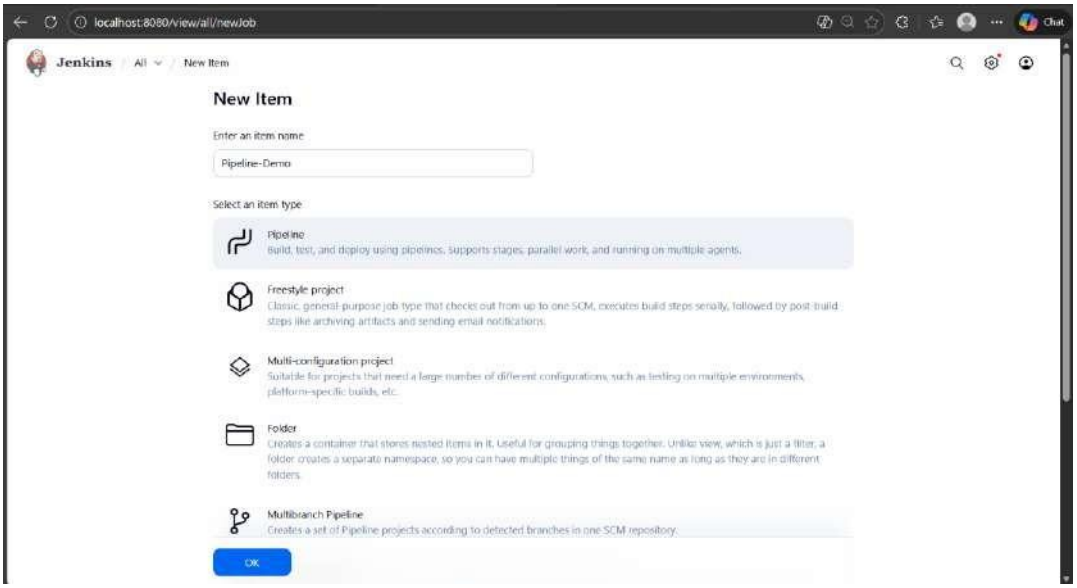
1. **What is Jenkins Pipeline?**

Jenkins Pipeline is a sequence of automated stages defined as code to perform build, test, and deployment tasks.

2. Scripted vs Declarative Pipeline

Feature	Scripted Pipeline	Declarative Pipeline
Syntax	Groovy-based	Structured & simple
Complexity	More flexible	Easy to learn
Error handling	Manual	Built-in
Recommended for	Advanced users	Beginners & labs

3. Create Pipeline Job



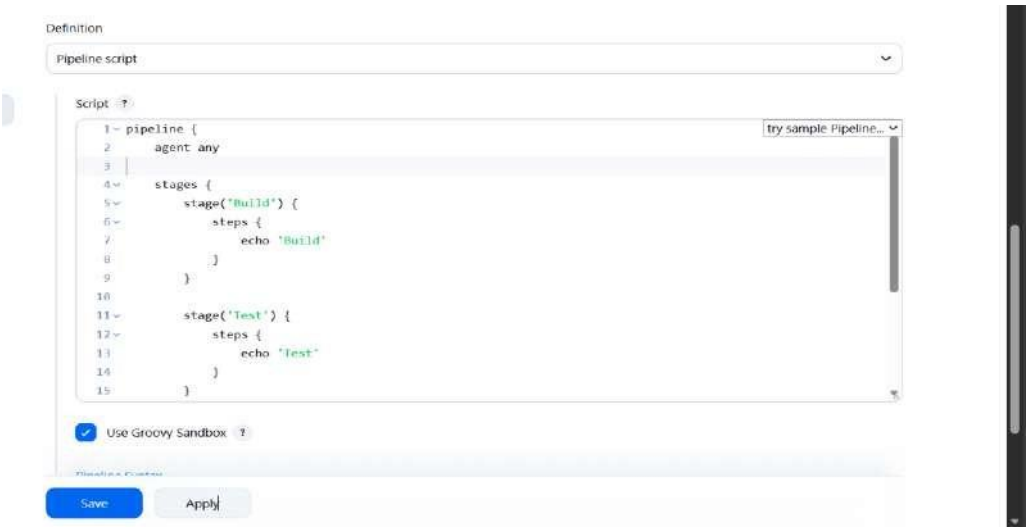
4. Write pipeline in Jenkins UI

5. **Add stages:**

Build

Test

Deploy



6. **Introduce:**

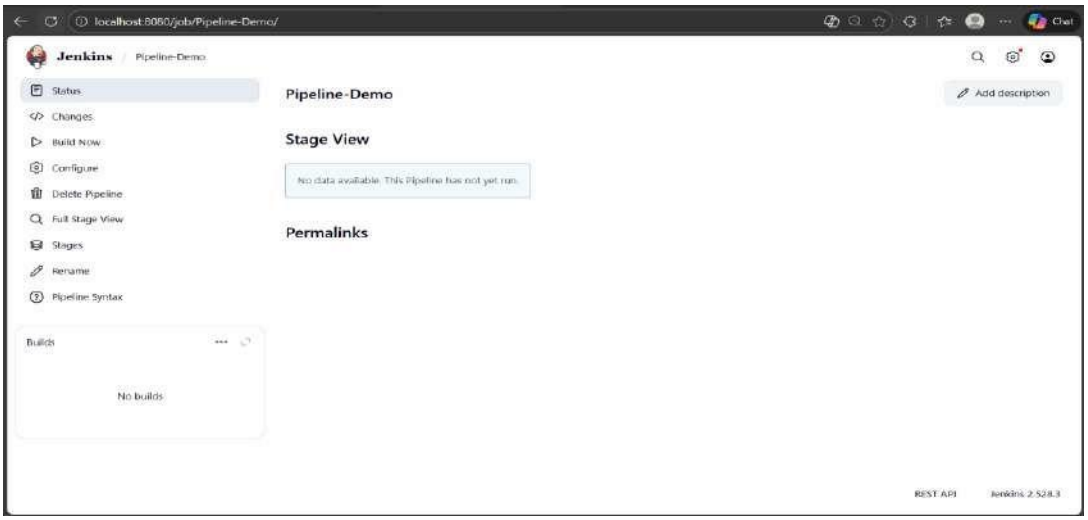
agent - Defines where the pipeline runs

stages - Logical grouping of pipeline phases

steps - Actual commands executed in a stage

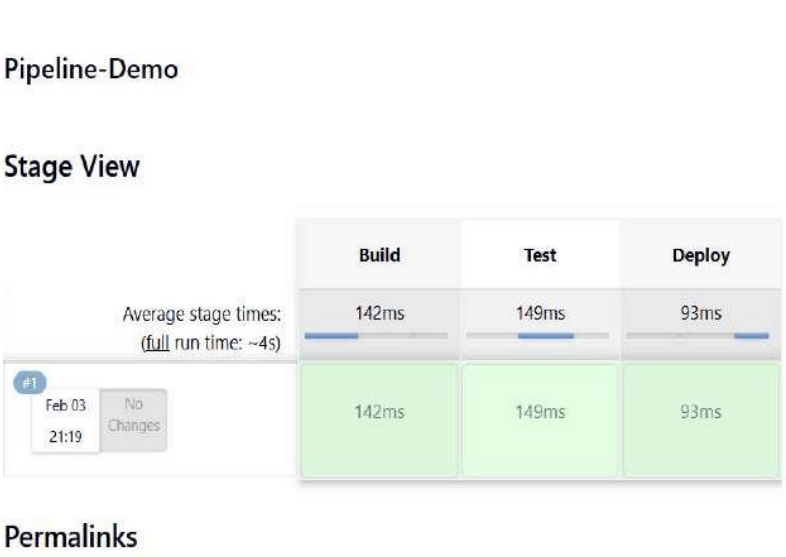
Run the Pipeline

1. Open Pipeline-Demo

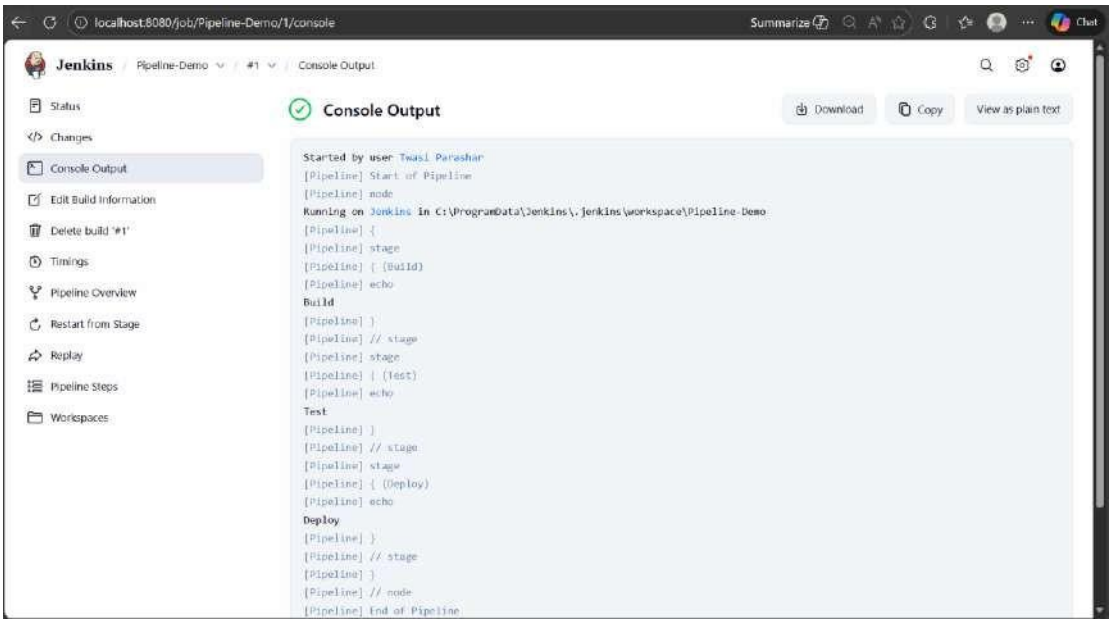


2. Click Build Now

3. Observe:
   - Stage View visualization
   - Build → Test → Deploy executed sequentially



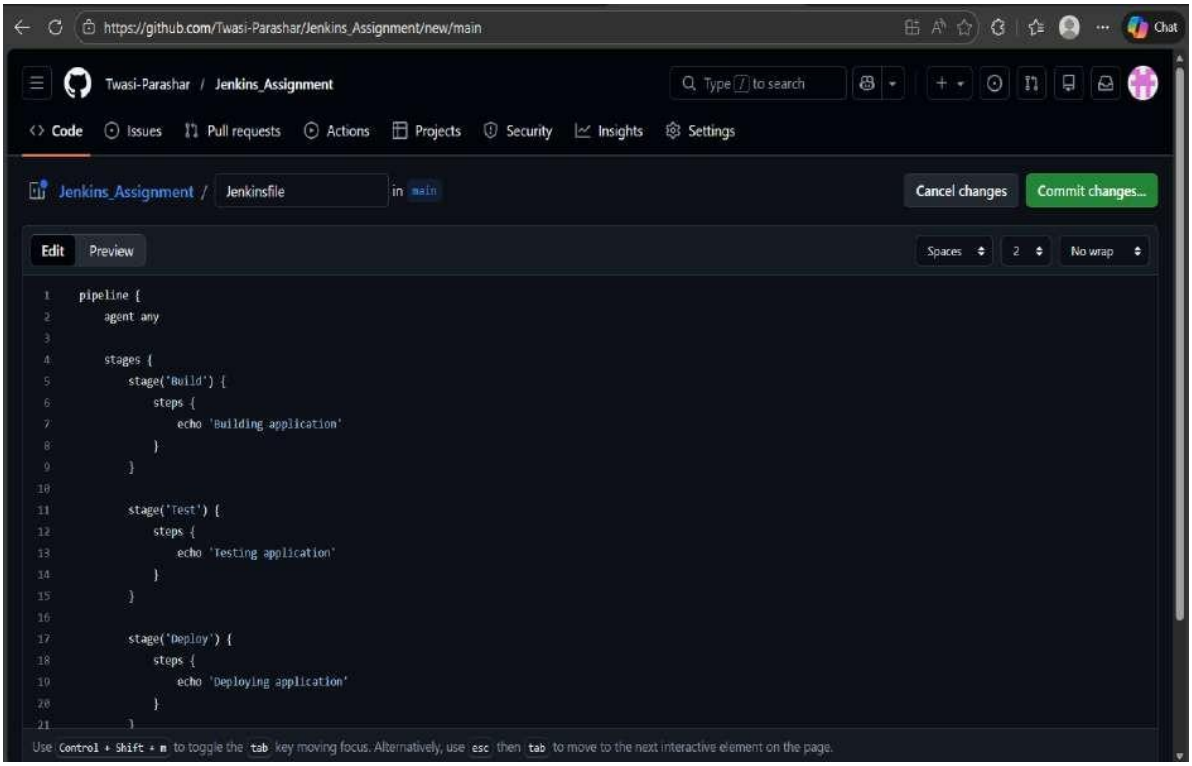
4. Click build number → **Console Output**



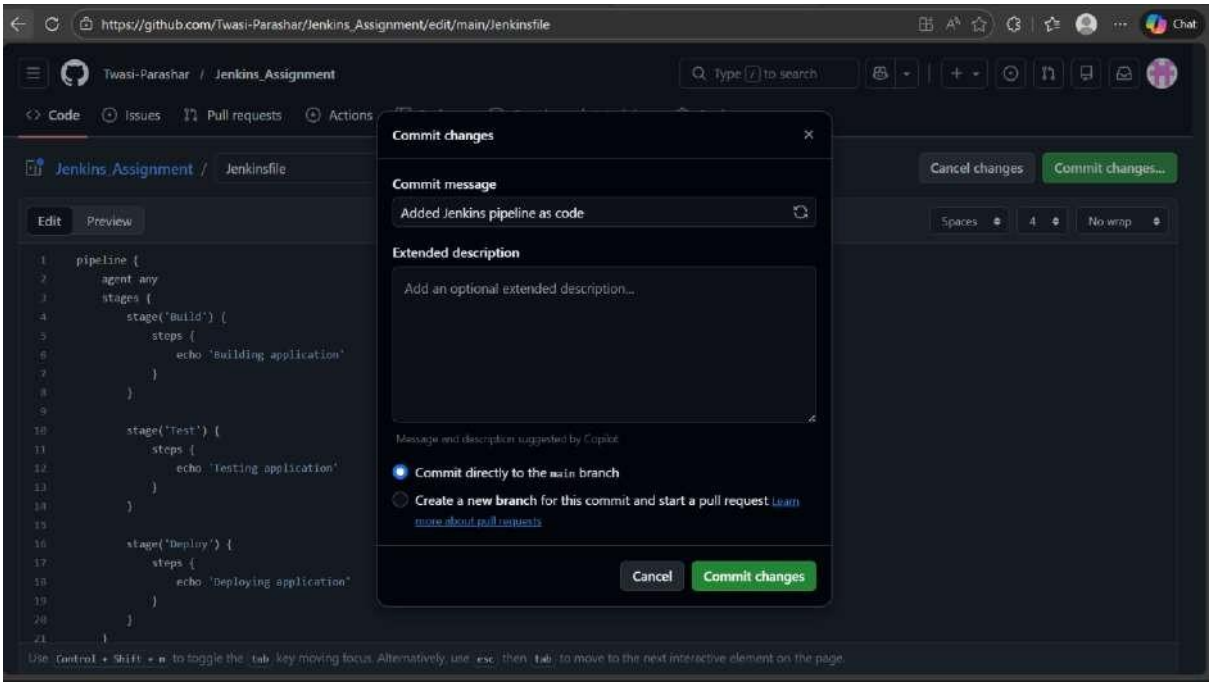
5. **Learning Goal:** Pipeline as code from repository

Stepwise Tasks

1. **Create Jenkinsfile**

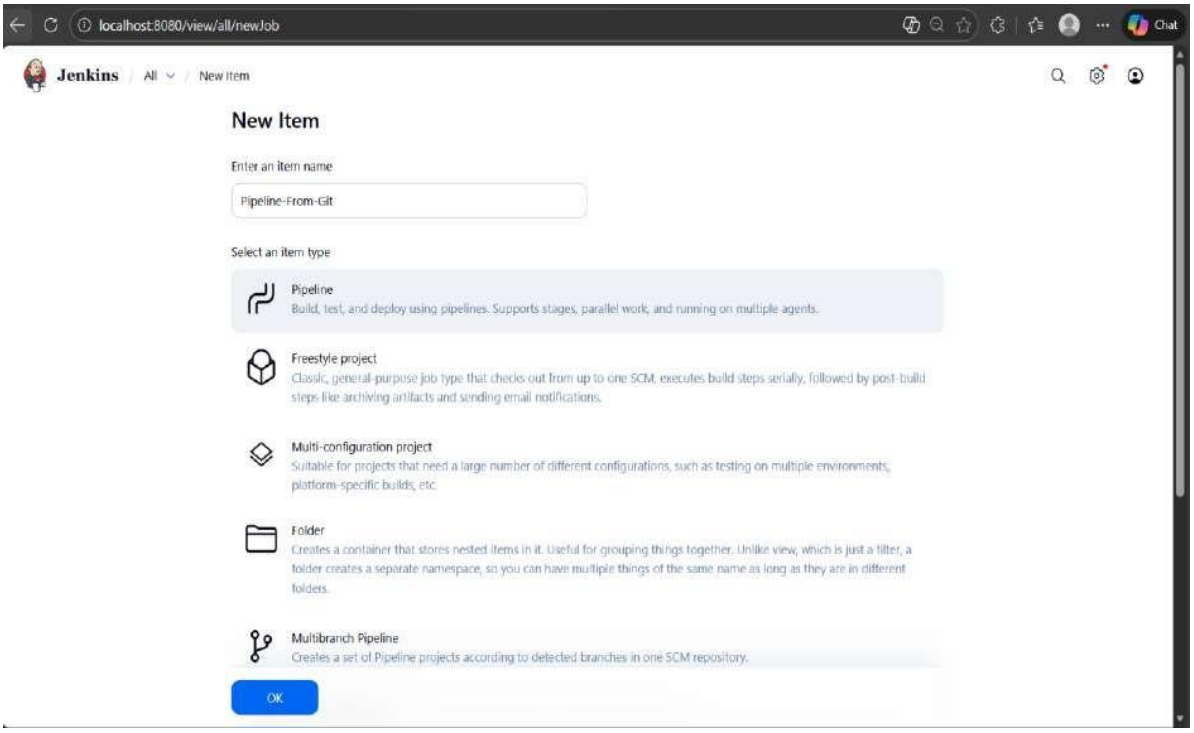


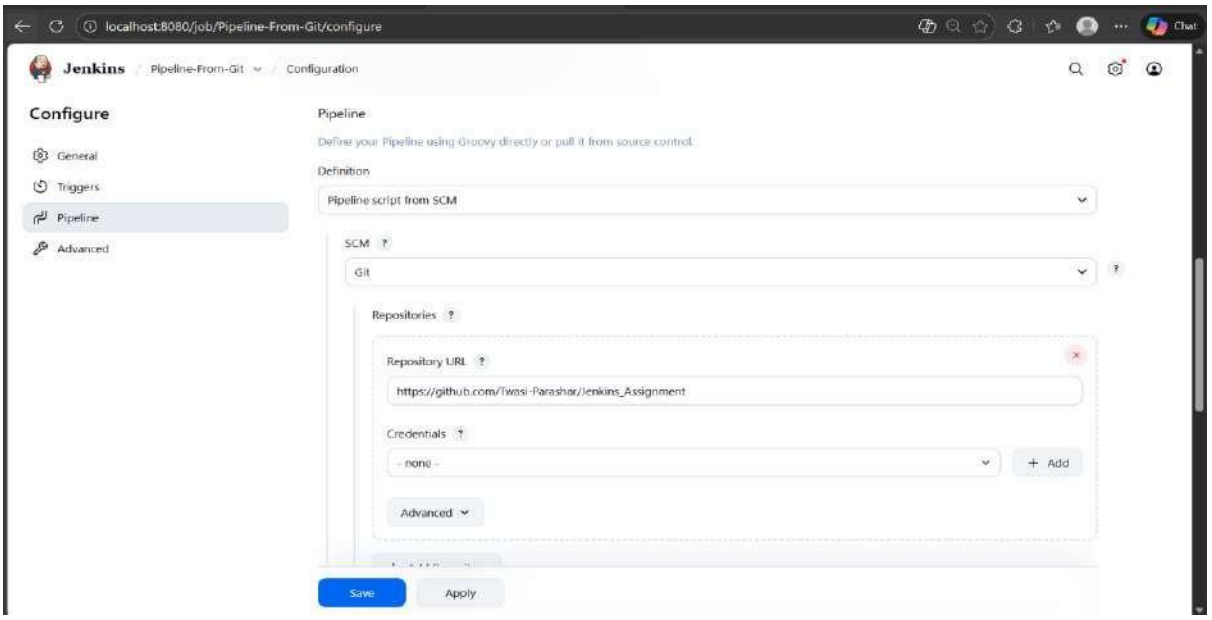
2. Commit Jenkinsfile to GitHub

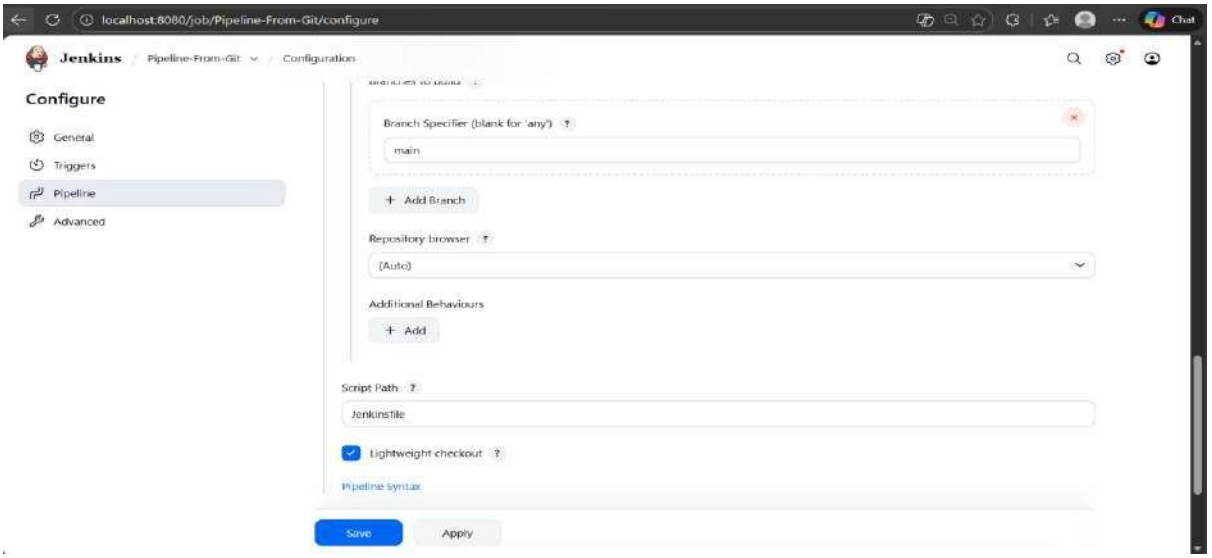


3. Configure Pipeline job:

SCM → Git

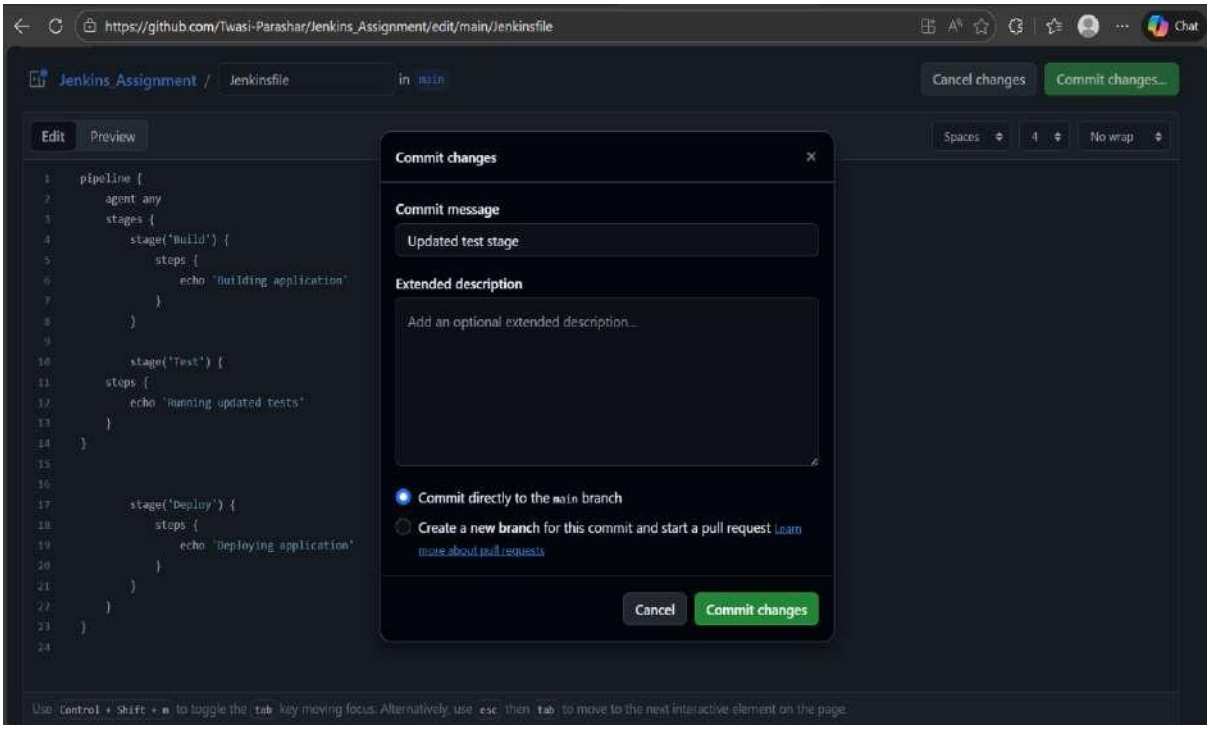


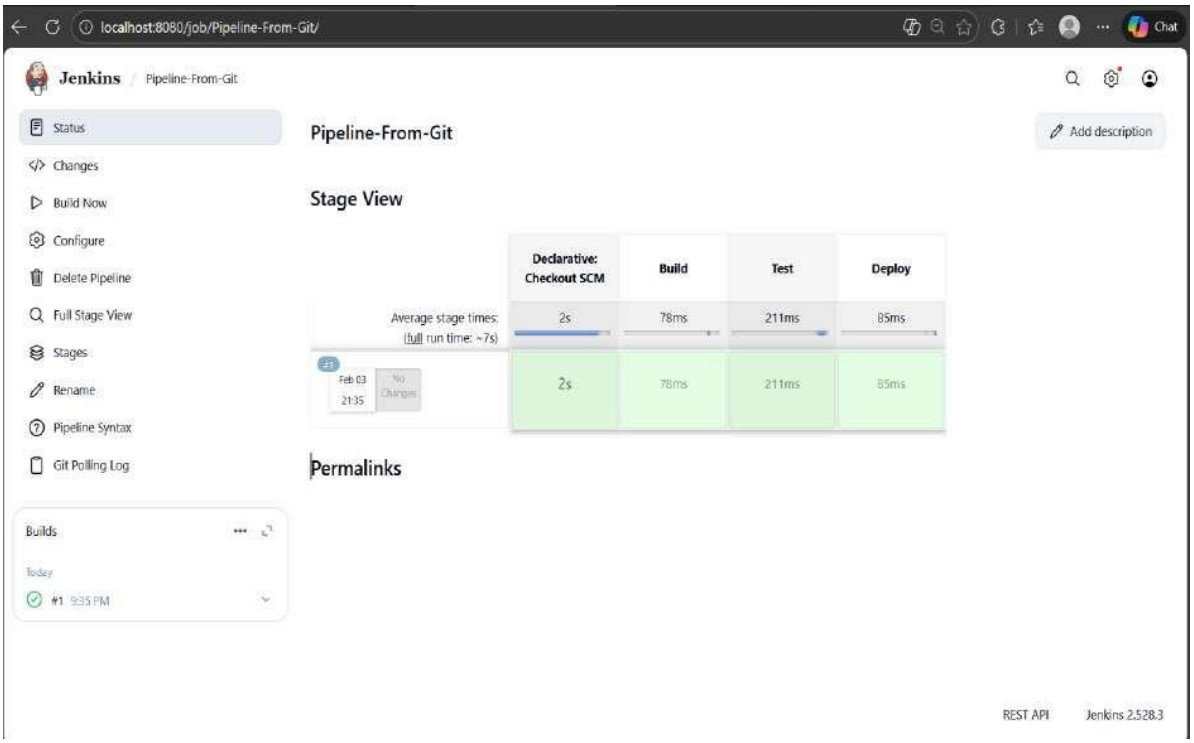




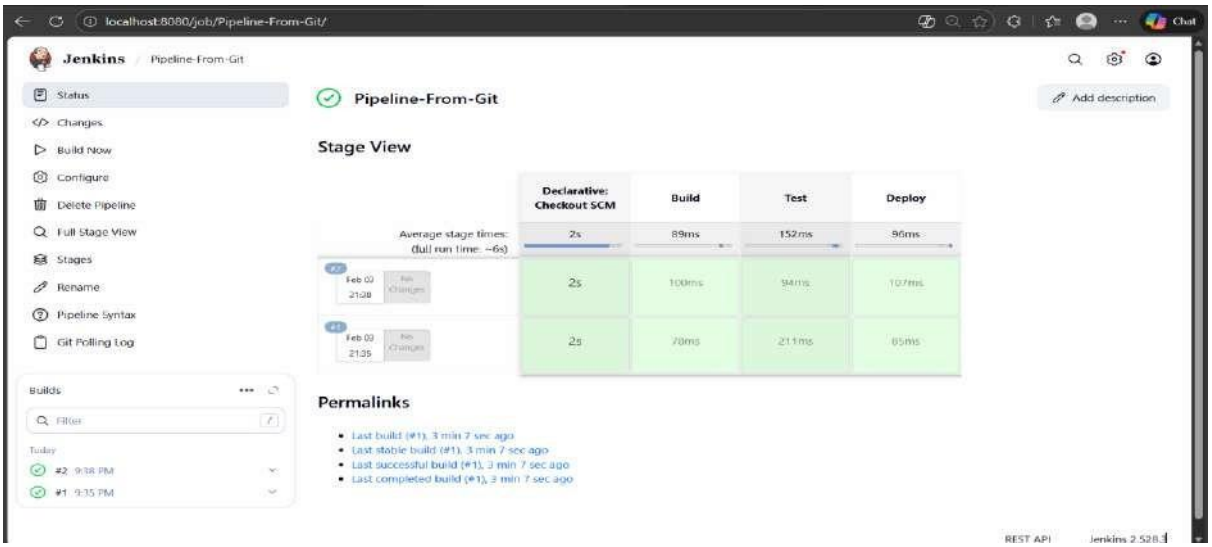
4. Trigger build automatically

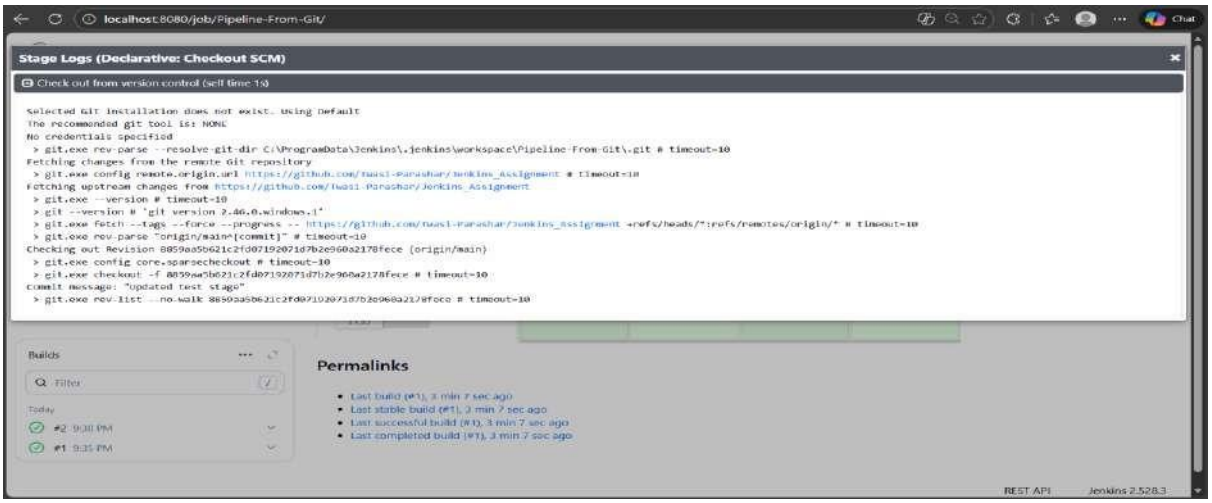
5. Modify Jenkinsfile and rebuild

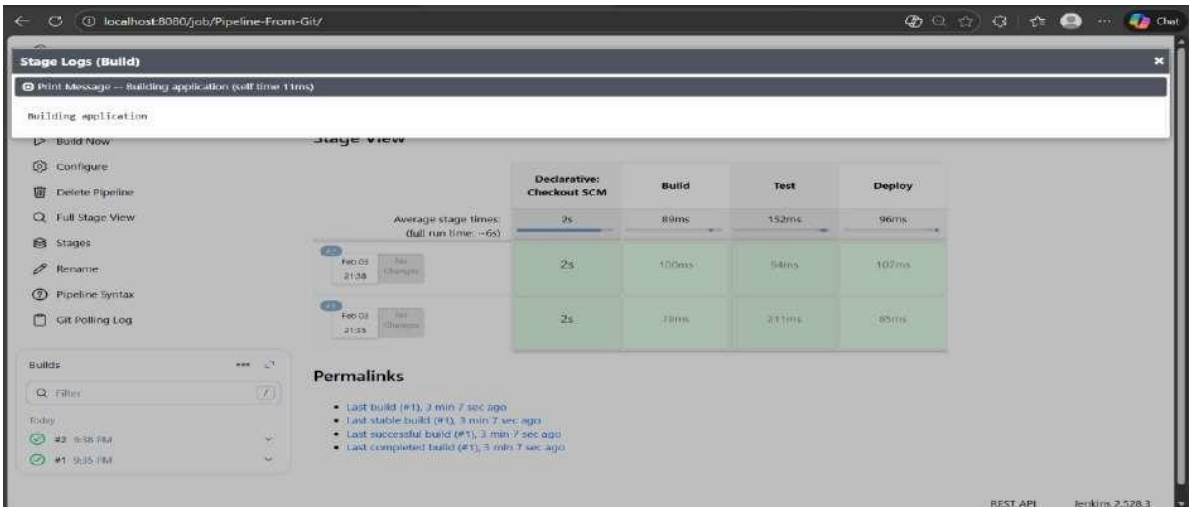


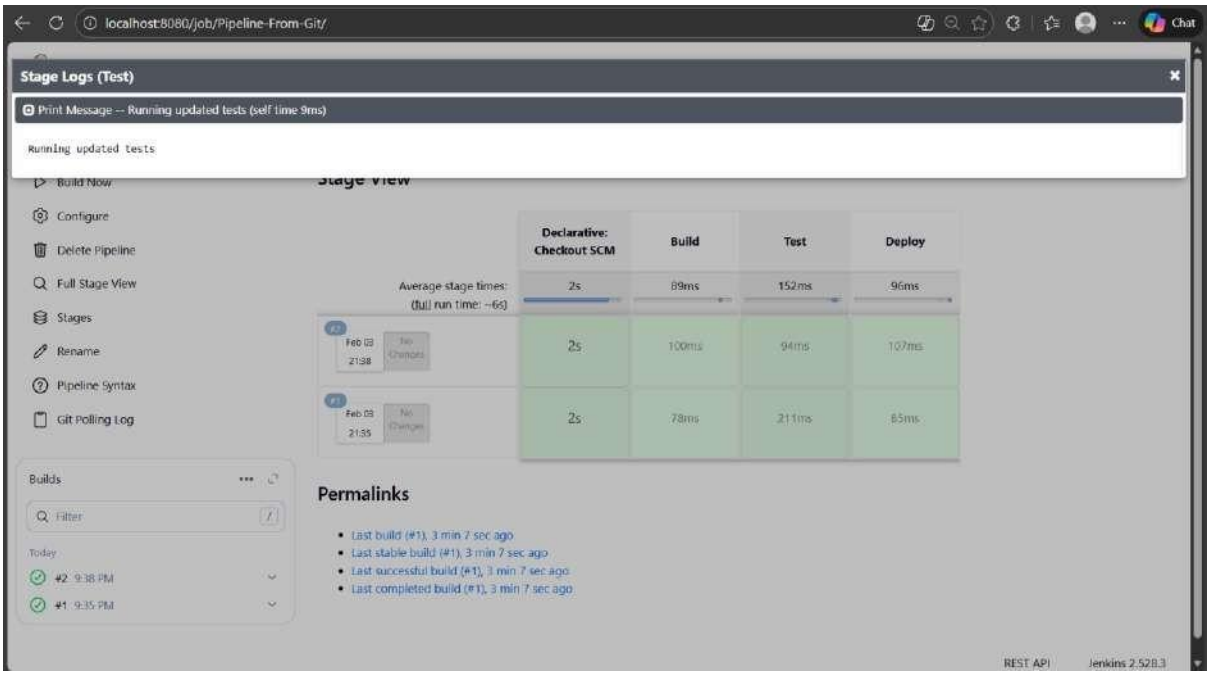


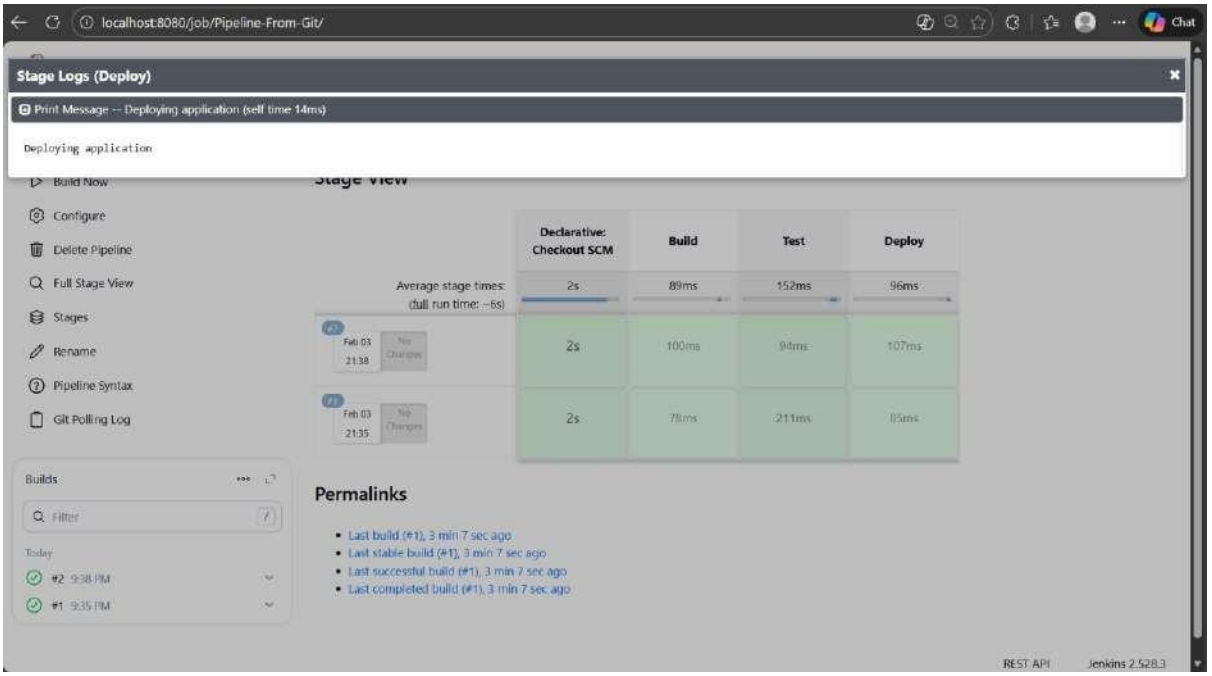
6. View Stage View








localhost:8080/job/Pipeline-From-Git/
Stage Logs (Test)
Print Message -- Running updated tests (self time 9ms)
Running updated tests
Build Now
Configure
Delete Pipeline
Full Stage View
Stages
Rename
Pipeline Syntax
Git Polling Log
Stage View
Declarative: Checkout SCM
Build
Test
Deploy
Average stage times: (full run time: ~6s)
2s
89ms
152ms
96ms
Feb 03 21:38
2s
100ms
94ms
107ms
Feb 03 21:35
2s
78ms
211ms
85ms
Builds
Filter
Today
#2 9:38 PM
#1 9:35 PM
Permalinks
Last build (#1), 3 min 7 sec ago
Last stable build (#1), 3 min 7 sec ago
Last successful build (#1), 3 min 7 sec ago
Last completed build (#1), 3 min 7 sec ago
REST API
Jenkins 2.528.3


localhost:8080/job/Pipeline-From-Git/
Stage Logs (Deploy)
Print Message -- Deploying application (self time 14ms)
Deploying application
Build Now
Configure
Delete Pipeline
Full Stage View
Stages
Rename
Pipeline Syntax
Git Polling Log
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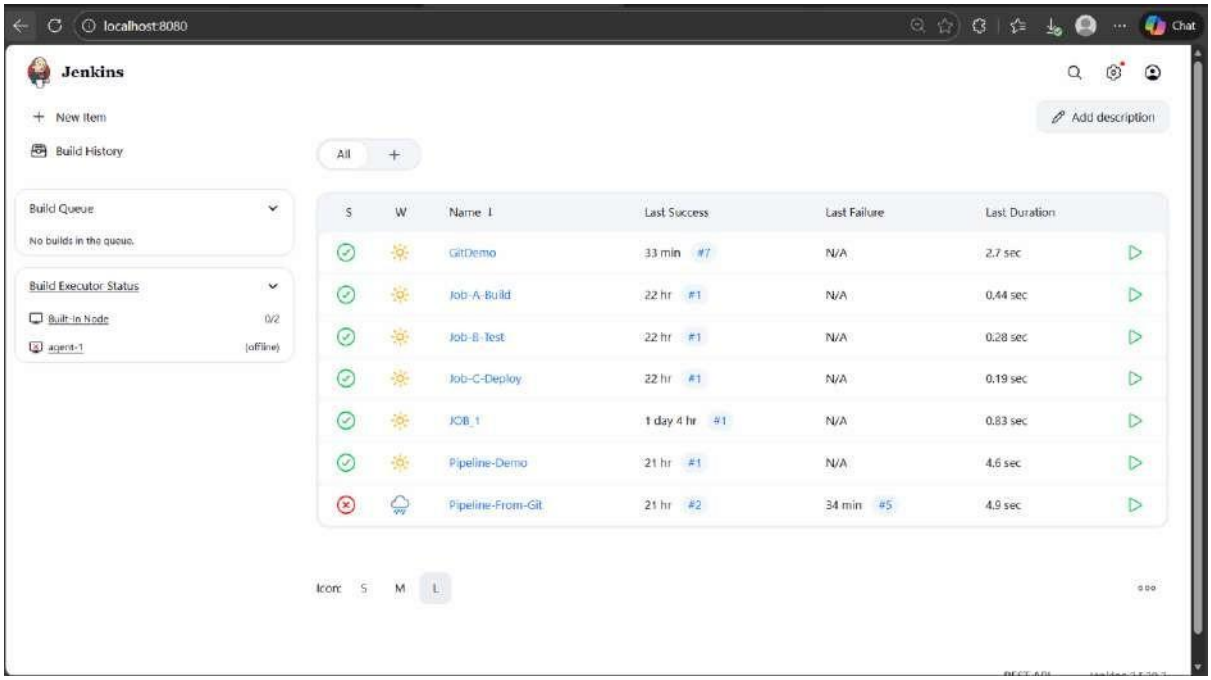
78ms
211ms
85ms
Builds
Filter
Today
#2 9:38 PM
#1 9:35 PM
Permalinks
Last build (#1), 3 min 7 sec ago
Last stable build (#1), 3 min 7 sec ago
Last successful build (#1), 3 min 7 sec ago
Last completed build (#1), 3 min 7 sec ago
REST API
Jenkins 2.528.3


localhost:8080
Chat
Jenkins
New Item
Build History
Add description
All
Build Queue
No builds in the queue.
Build Executor Status
Built-In Node 0/2
agent-1 (offline)
S W Name ↓ Last Success Last Failure Last Duration
GitDemo 33 min #7 N/A 2.7 sec
Job-A-Build 22 hr #1 N/A 0.44 sec
Job-B-Test 22 hr #1 N/A 0.28 sec
Job-C-Deploy 22 hr #1 N/A 0.19 sec
JOB_1 1 day 4 hr #1 N/A 0.83 sec
Pipeline-Demo 21 hr #1 N/A 4.6 sec
Pipeline-From-Git 21 hr #2 34 min #5 4.9 sec
Icon: S M L